

A Systematic Comparison of Six Machine Learning Algorithms for Binary Heart Disease Classification Using Multi-Source Clinical Data

Project Report

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A Systematic Comparison of Six Machine Learning Algorithms for Binary Heart Disease Classification Using Multi-Source Clinical Data

Abstract

Cardiovascular disease is the most common cause of global mortality. It is a prime reason for an approximate annual death toll of 17.9 million individuals. Early and accurate diagnoses are essential. But, as physiology slowly declines until the onset of a significant cardiac event, standard clinical assessments fail to detect patients. This research establishes a machine learning framework to enable non-invasive heart disease screening with the use of a single clinical dataset (917 patient records) gained from five different cardiovascular databases. Six classification algorithms were evaluated: Logistic Regression, Random Forest, Naive Bayes, Support Vector Machine, Gradient Boosting algorithms and Multilayer Perceptron algorithms. A preprocessing pipeline that addressed biologically implausible zero-values within the Cholesterol column via KNN imputation and applied per-fold StandardScaler normalisation through sklearn Pipelines was constructed.

For optimal Hyper-parameter selection, tuning methods were employed using GridSearchCV based on F1-score. Final evaluations completed under stratified cross-validation (10-fold) and five performance metrics; Accuracy, Precision, Recall, F1-Score, and AUC-ROC are used for model evaluation. Gradient Boosting achieved the highest F1-score of 0.8815 and AUC of 0.9262.

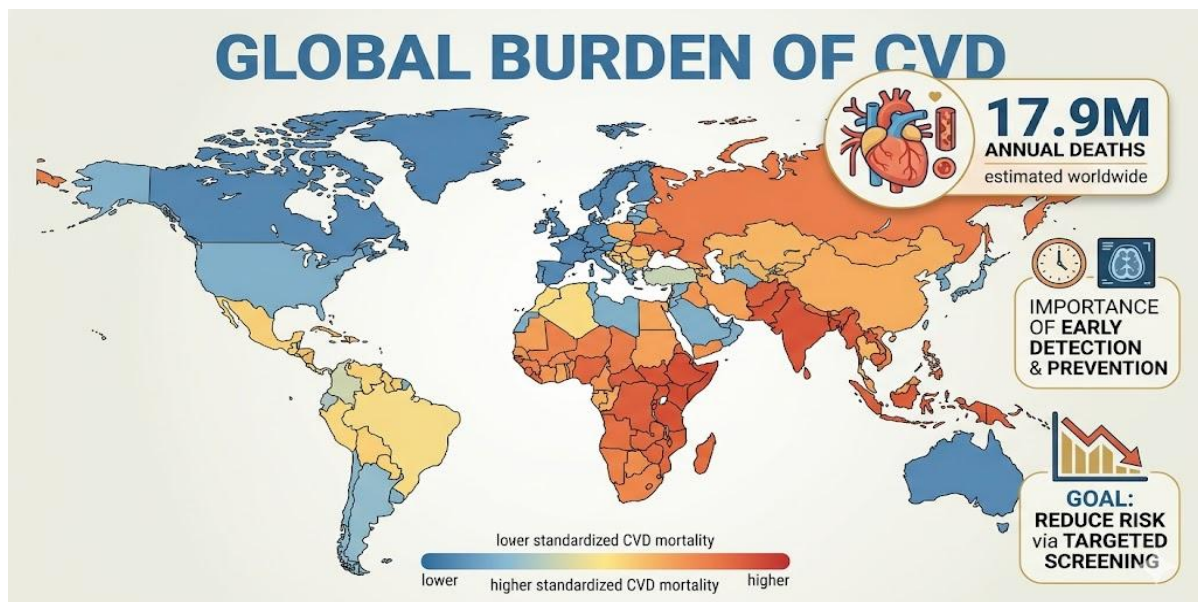
Though the performance spread across all six tuned models spanned only 1.09 percentage points in accuracy, a convergence confirmed as statistically non-significant by both paired t-tests and McNemar's tests. This finding indicates that on structured clinical tabular data of this scale, feature quality governs the performance ceiling more than algorithmic complexity. Feature importance analysis identified STSlope, Oldpeak, ChestPainType, and MaxHR as the dominant predictors across models, consistent with established cardiac physiology. Given the absence of statistically distinguishable differences, Logistic Regression emerges as the most operationally compelling candidate for clinical deployment, offering equivalent predictive performance alongside full coefficient-level interpretability, a property of increasing regulatory and clinical significance in medical decision support.

Keywords: Heart disease prediction, machine learning, KNN imputation, Gradient Boosting, Logistic Regression, clinical decision support, cardiovascular risk classification

1. Introduction

1.1 Clinical Background and Objective

Cardiovascular disease is the single largest cause of global mortality. According to the World Health Organization (2024), 17.9 million deaths are attributed to cardiac conditions annually, accounting for roughly 32% of all global deaths. A core clinical challenge is that many patients progress through an asymptomatic window where their physiological markers are deteriorating but remain within ranges that standard examination may not flag as abnormal. Liu et al., (2025) note that this delayed detection window is one of the primary reasons that preventable cardiac events continue to occur even in patients with access to routine healthcare.



Machine learning offers a practical route to closing this gap. By learning from the recorded clinical measurements of previously diagnosed patients, a trained classifier can evaluate new patient profiles and flag those whose feature combinations align with known disease patterns. This study builds and evaluates six such classifiers on a multi-source clinical dataset, with the goal of identifying which algorithm provides the most reliable and clinically safe predictions for heart disease diagnosis.

1.2 Dataset Description

The dataset used in this study is the Heart Failure Prediction Dataset, compiled by Fedesoriano (2021) and available at <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

It brings together patient records from five independent cardiovascular databases: Cleveland, Hungary, Switzerland, Long Beach VA, and the Statlog Heart Disease collection. In total, the dataset contains 917 records, 11 clinical input features after cleaning. The binary target variable is the HeartDisease, where a value of 1 indicates a cardiac condition and 0 indicates no disease. Table 1 lists each feature along with its type and clinical meaning.

Table 1: Feature descriptions of the Heart Failure Prediction Dataset

Feature	Type	Description
Age	Numeric	Patient age in years
Sex	Categorical	Biological sex (M = Male, F = Female)
ChestPainType	Categorical	ASY = Asymptomatic, ATA = Atypical Angina, NAP = Non-Anginal Pain, TA = Typical Angina
RestingBP	Numeric	Resting blood pressure (mm Hg)
Cholesterol	Numeric	Serum cholesterol (mg/dl)
FastingBS	Binary	Fasting blood sugar > 120 mg/dl (1 = yes, 0 = no)
RestingECG	Categorical	Resting ECG result: Normal, ST, or LVH
MaxHR	Numeric	Maximum heart rate achieved during exercise
ExerciseAngina	Categorical	Exercise-induced angina (Y = yes, N = no)
Oldpeak	Numeric	ST depression induced by exercise relative to rest
STSlope	Categorical	Slope of the peak exercise ST segment: Up, Flat, or Down

1.3 Exploratory Data Analysis

1.3.1 Class Distribution

The class distribution of the HeartDisease target variable is shown in Figure 1. Out of a total of 507 positive cases (55.3%) and 410 negative cases (44.7%), there is an approximate 55:45 ratio of imbalance. The dataset's distribution is sufficiently close to the balance point that standard evaluation metrics will remain valid indicators of model performance (Accuracy & F1-Score) without the need for synthetic oversampling. Model training will also have occurred with a representative sample of patients that have both positive and negative disease diagnoses in terms of the model accuracy given the medical relevance of the balanced dataset.

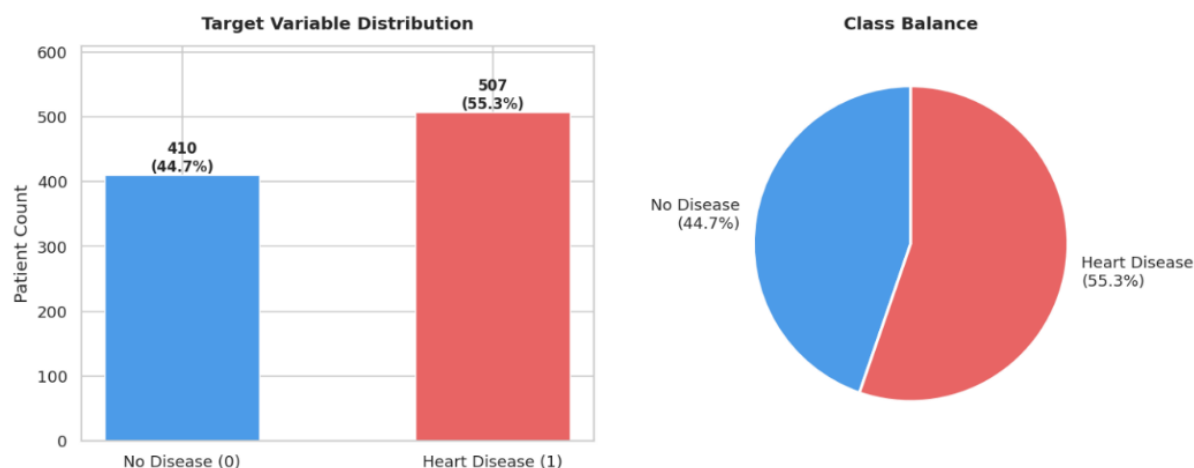


Figure 1: Analysis of Target Variable (HeartDisease) Distribution

1.3.2 Numeric Feature Distributions

Figure 2 shows overlapping histograms for the five continuous features, stratified by diagnosis. Age and MaxHR show clear distributional shifts between the two classes. There is a significant increase in heart disease from the age of approximately 50 to 60. Individuals who do not have heart disease, often will have a much greater maximum heart rate than those with heart disease. Oldpeak is an indicator with a very clear distinction between the two groups; healthy individuals have values very dense around zero, while individuals who are positive for heart disease have all ranges of values above zero, confirming this variable is the most reliable predictor of heart disease. Cholesterol and RestingBP show more overlap between classes, indicating they are secondary rather than primary drivers of classification.

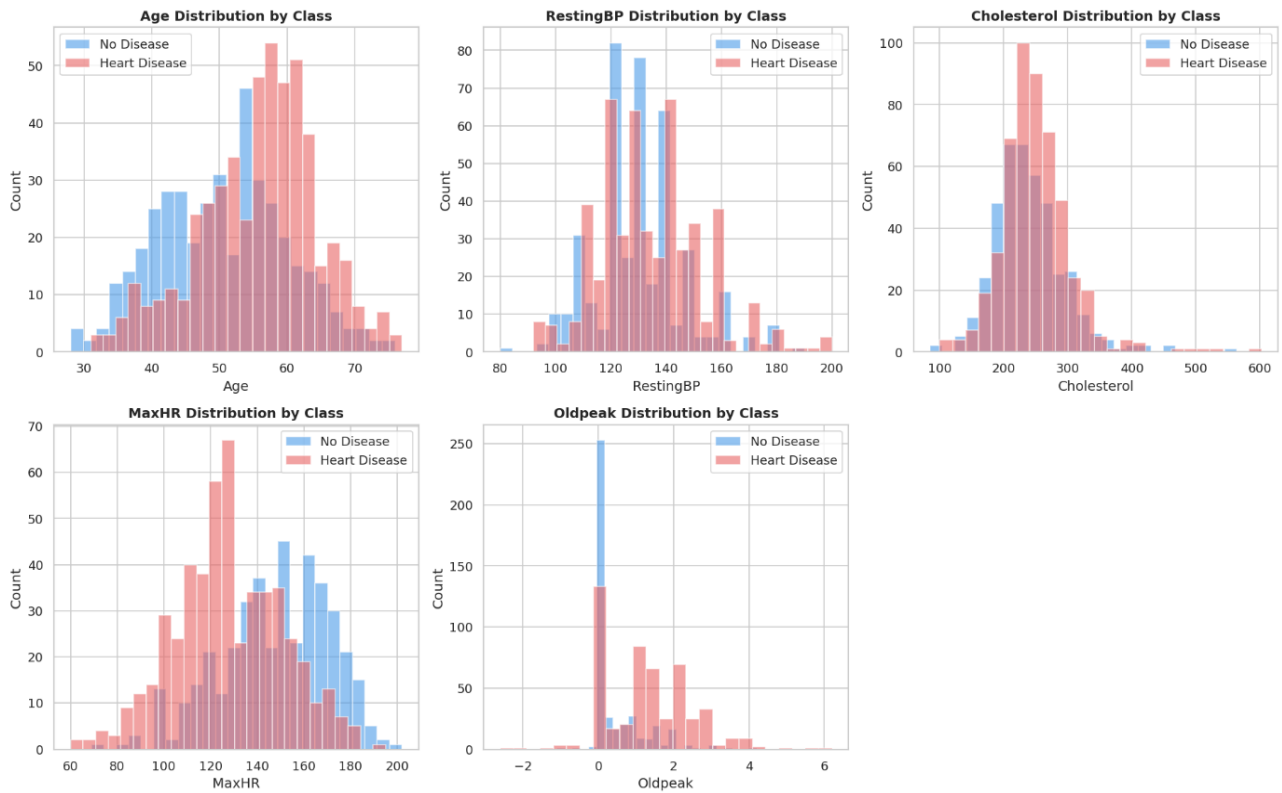


Figure 2: Comparative Analysis of Numeric Feature Distributions

1.3.3 Categorical Feature Analysis

Figure 3 presents cross-tabulated bar charts for all six categorical features. ChestPainType shows the starkest contrast, with Asymptomatic (ASY) patients carrying the highest disease prevalence while those with Atypical Angina (ATA) are predominantly healthy. ExerciseAngina follows a similar pattern, where exercise-induced chest pain (Y) functions as a near-definitive indicator of cardiac risk in this dataset. STSlope reinforces this further, with a Flat slope heavily associated with disease presence and an Up slope characteristic of healthy patients. FastingBS shows that patients with blood sugar above 120 mg/dl carry a meaningfully higher proportion of positive diagnoses, validating its inclusion as a secondary risk marker. On the demographic side, male patients show a higher frequency of heart disease within this clinical sample. RestingECG is perhaps the most clinically instructive finding: even patients with a Normal ECG result can carry heart disease, which directly supports the need for multi-feature machine learning classifiers over any single diagnostic marker.

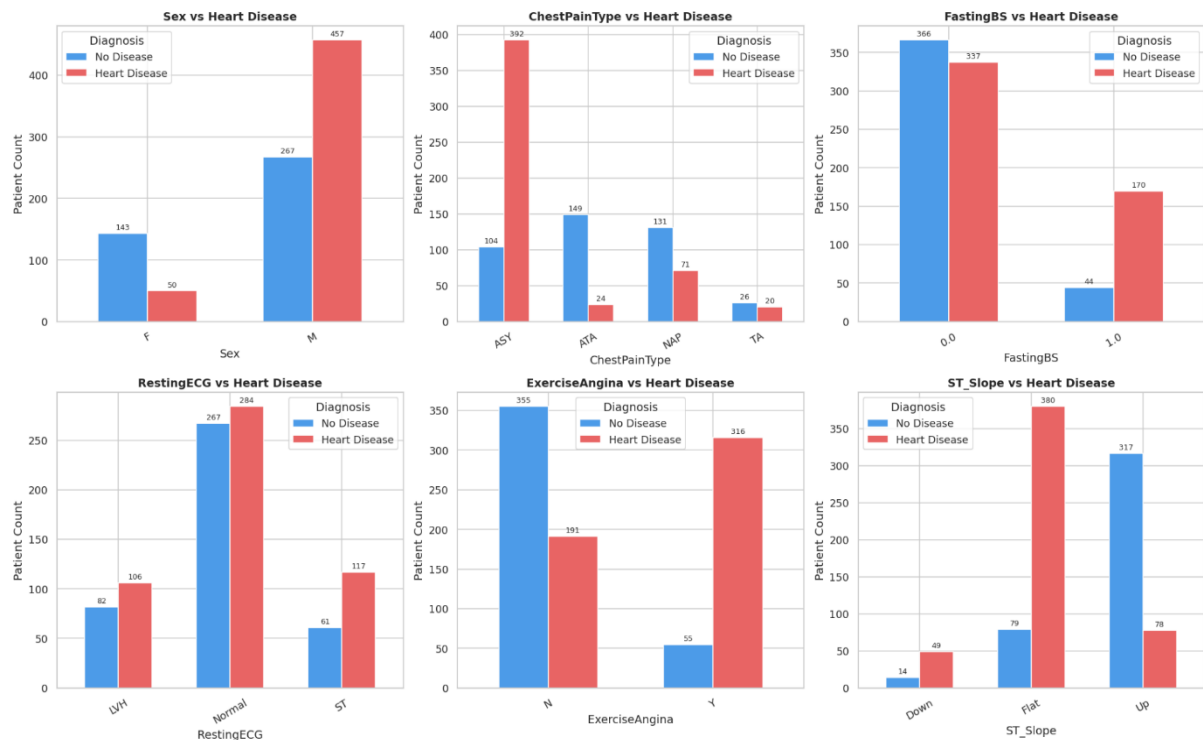


Figure 3: Categorical Feature Correlation with Heart Disease Diagnosis

1.3.4 Age Group Risk Profiling

Figure 4 bins the continuous Age variable into four cohorts and shows disease prevalence within each group. The risk profile rises sharply with age:

Table 2: Heart Disease Risk by Age

Age Group	Disease Rate	Patient Count
Under 40	32.5%	80
40-49	40.3%	211
50-59	56.6%	373
60+	73.1%	253

The 50-59 group contains the largest patient volume, and the 60+ group carries the highest disease rate. Even the youngest cohort shows a 32.5% disease rate, suggesting clinical monitoring should not be deferred until late middle age.

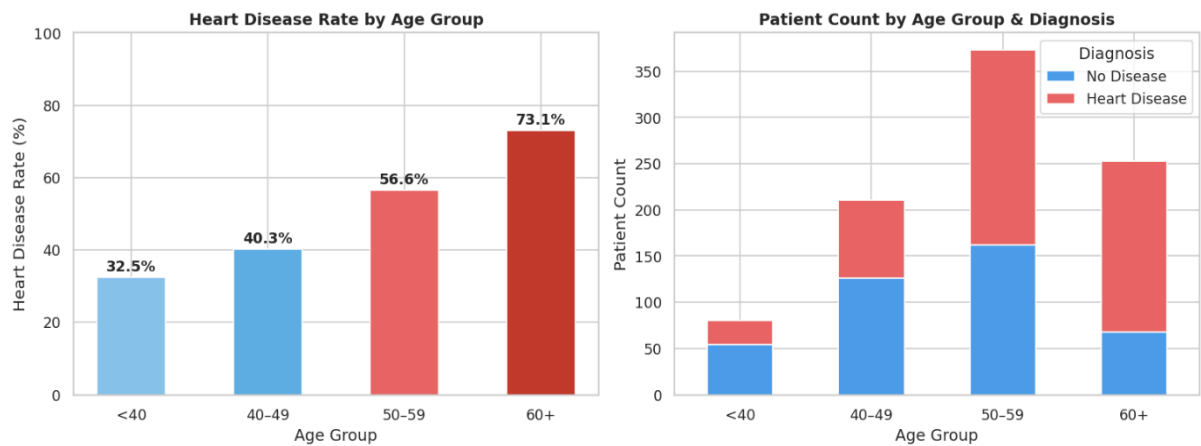


Figure 4: Age Group Risk Profiling and Demographic Trends

1.3.5 MaxHR vs Age Scatter Analysis

Figure 5 plots MaxHR against Age for both cohorts with linear regression trend lines overlaid. Both groups show a negative correlation between age and peak heart rate, consistent with normal physiological ageing. However, disease-positive patients maintain consistently lower MaxHR values across all age brackets compared to healthy patients at the same age. This gap provides an additional decision boundary for classifiers, particularly ensemble methods and SVM which can exploit this interaction across the feature space.

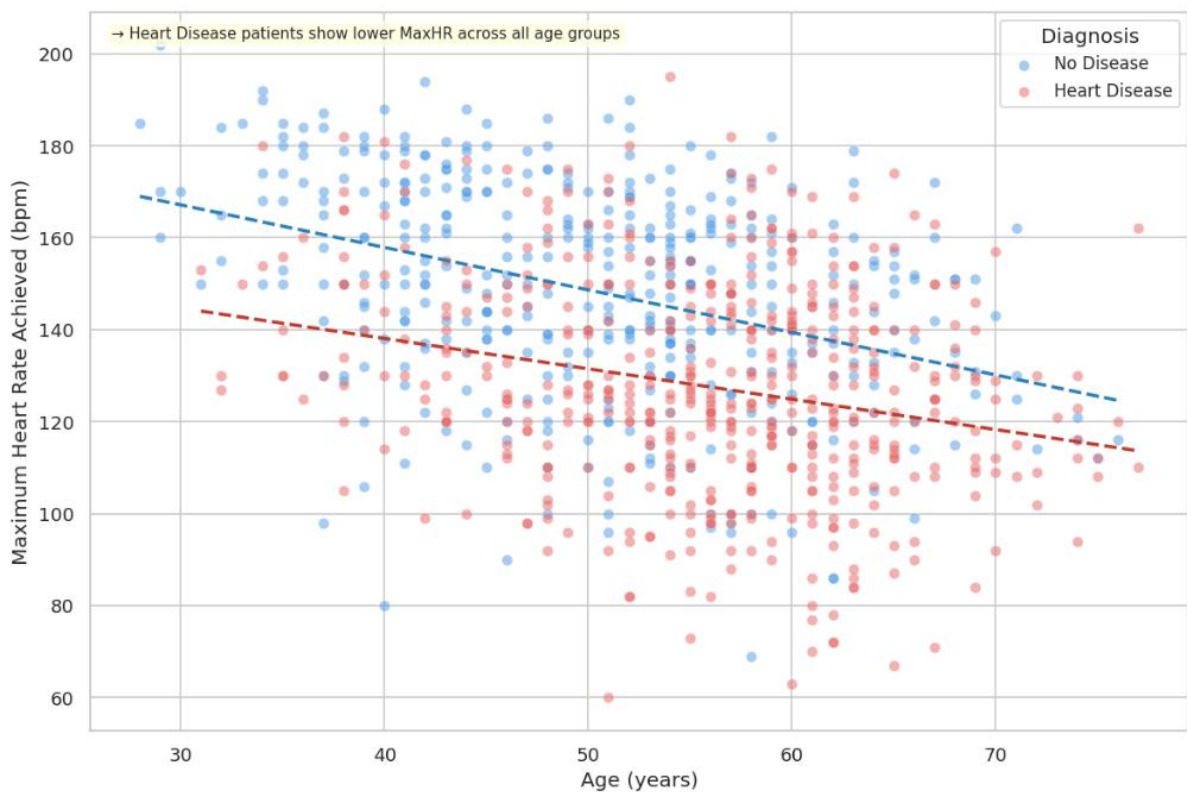


Figure 5: Correlation Analysis of Maximum Heart Rate and Age

1.3.6 Correlation Heatmap

Figure 6 presents a triangular correlation matrix for all numeric features. HeartDisease shows its strongest correlations with Oldpeak (positive) and MaxHR (negative). Feature-to-feature correlations are generally below 0.70, with the moderate Age-MaxHR link being the most notable. This confirms that the feature set does not carry significant redundancy and that multicollinearity is unlikely to destabilise linear model coefficients, though VIF testing was conducted to verify this formally.

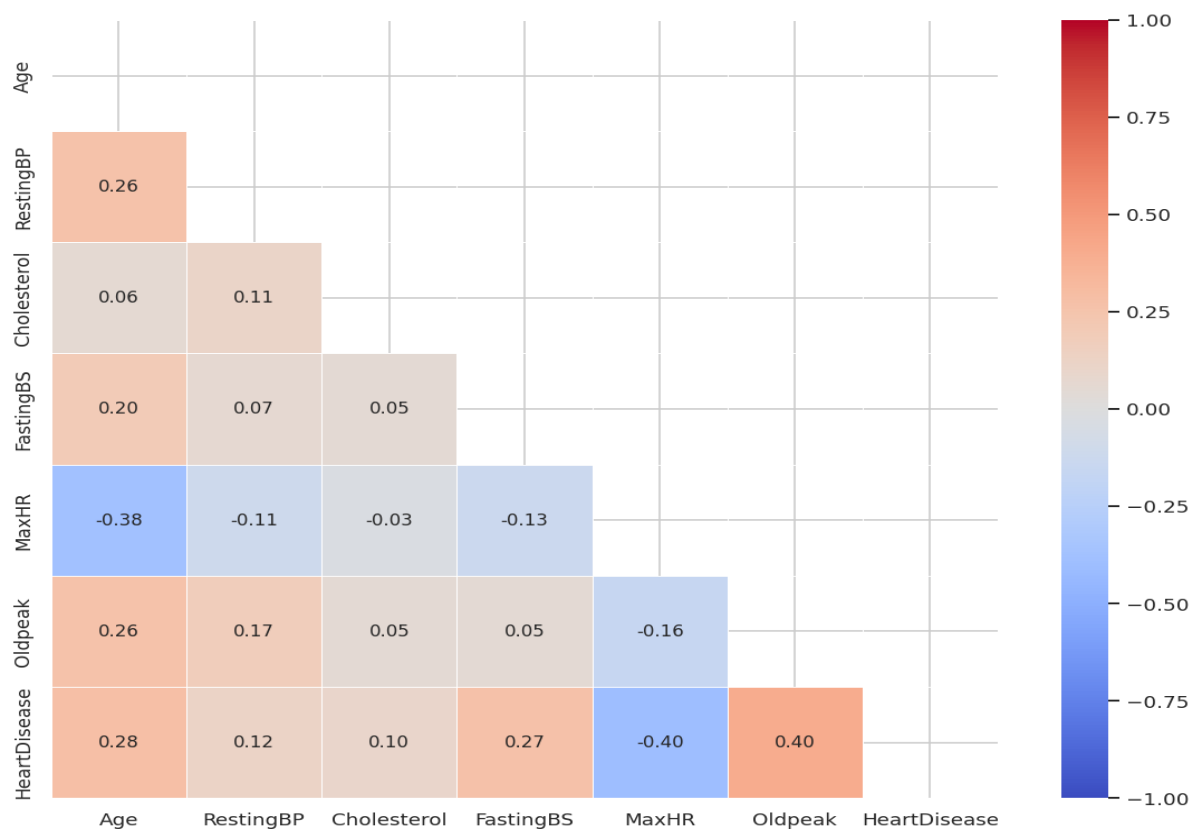


Figure 6: Correlation Heatmap of Numeric Physiological Features

1.3.7 Point-Biserial Correlation

Figure 7 quantifies the statistical association between each continuous feature and the binary target. Results are:

All five features achieved $p < 0.05$, confirming their inclusion in the model training is statistically justified. The near-equal and opposite coefficients for Oldpeak and MaxHR confirm these two features as the dominant numeric predictors.

Table 3: Feature Correlation

Feature	Correlation (r)	Significant
Oldpeak	+0.404	Yes
MaxHR	-0.401	Yes
Age	+0.282	Yes
RestingBP	+0.118	Yes
Cholesterol	+0.095	Yes

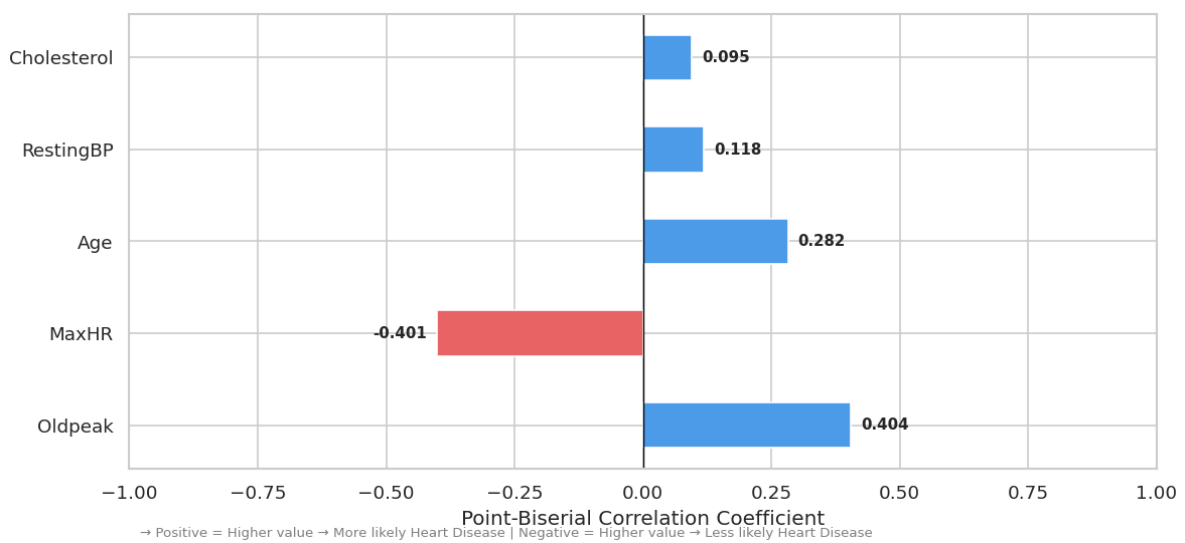


Figure 7: Point-Biserial Correlation Analysis of Numeric Predictors

1.3.8 STSlope and Oldpeak Interaction Analysis

Figure 8 plots mean Oldpeak values across STSlope categories, stratified by diagnosis. Patients with Flat or Down slopes combined with elevated Oldpeak show a near-definitive disease signature. For disease-positive patients with a Flat slope the mean Oldpeak is 1.57, whereas for healthy patients with the same slope it is 0.55. For Down slopes, this gap widens further: 2.37 for disease-positive versus 0.89 for healthy patients. Patients with an Up slope show lower Oldpeak regardless of diagnosis, but even within this group, disease-positive patients maintain a higher mean (0.59) than healthy ones (0.24). This interaction explains why tree-based models

perform strongly: they can capture the conditional importance of Oldpeak depending on STSlope state, something a simple linear boundary cannot express.

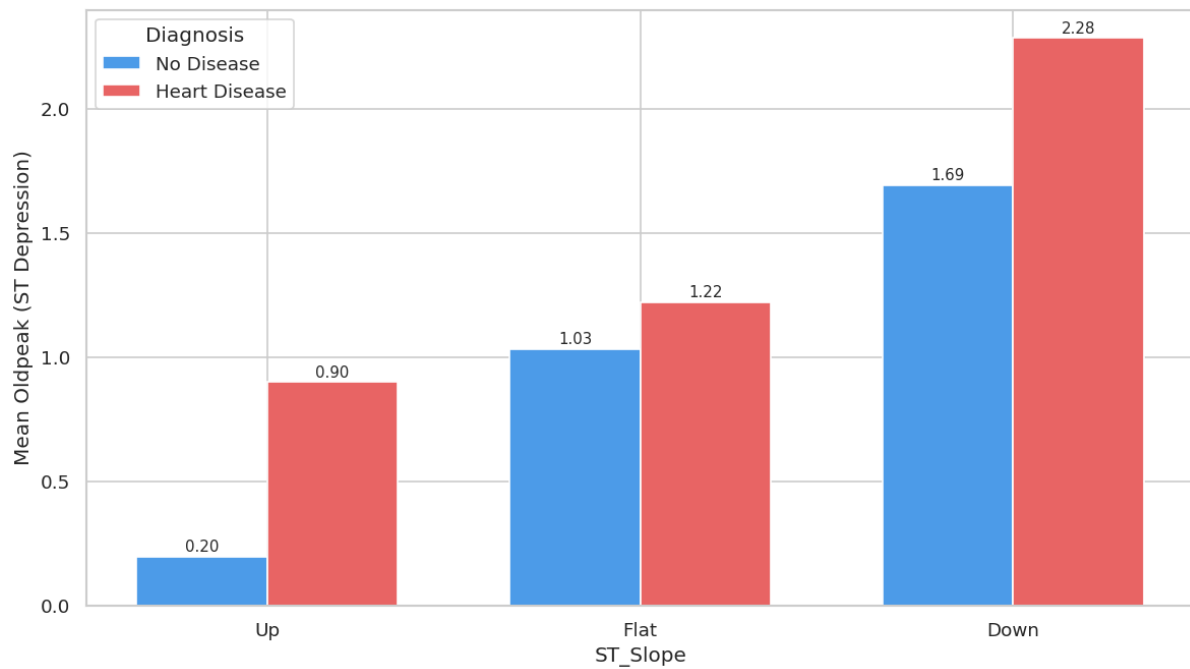
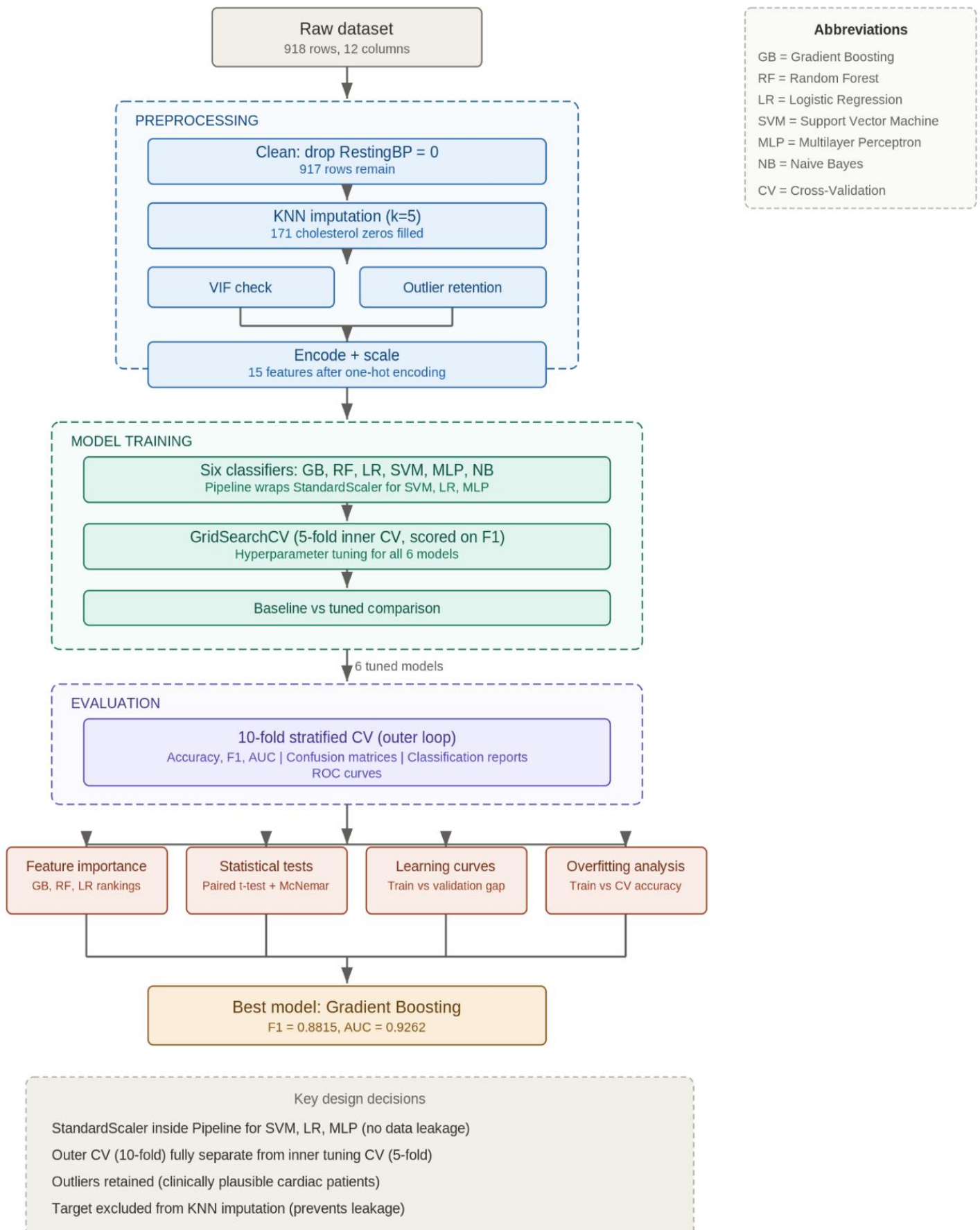


Figure 8: Interaction Analysis: Mean Oldpeak by ST_Slope and Diagnosis

2. Methodology



2.1 Data Wrangling

2.1.1 Data Cleaning

The raw dataset contained 918 records across 11 features. Among them one record was identified with a RestingBP value of 0, which is physiologically impossible. So, it was removed, reducing the working dataset to 917 patient records. Cholesterol values coded as 0 were treated as missing values because it is not likely to be zero clinically, therefore all 171 entries (18.6% of the cleaned dataset) were kept in the dataset and marked for imputation in the next step instead of being discarded, so that the entire patient population could be retained for the training of the model.

2.1.2 Missing Value Imputation

The 171 missing cholesterol values (18.6% of the cleaned set) were filled using KNN imputation with $k = 5$. The target variable was excluded from the neighbour search to avoid leakage. After imputation, cholesterol had a mean of 245.0 and standard deviation of 54.7, which sat within range of the non-missing values, so the imputation did not pull the distribution in any particular direction. Lin and Tsai (2020) tested several imputation strategies on clinical data with missing rates under 30% and found KNN to be the most stable.

Table 4: Cholesterol Statistics Before and After KNN Imputation ($K=5$, Target Excluded)

Statistic	Before Imputation	After KNN Imputation
Records with values	746 (81.4%)	917 (100%)
Missing (coded as 0)	171 (18.6%)	0 (0%)
Mean (mg/dL)	244.6	245.0
Std Deviation	59.2	54.7

2.1.3 Outlier Retention

IQR screening flagged 27 outliers in RestingBP, 28 in Cholesterol, 16 in Oldpeak, and 2 in MaxHR. Age had none. All outliers were kept. A resting BP of 170 or cholesterol above 400 is unusual statistically but not clinically, these are cardiac patients, and removing their extreme readings would trim the very population the model needs to learn from. Gradient Boosting and Random Forest split on rank order rather than raw values, so extreme readings do not distort their boundaries the way they would in a distance-based model.

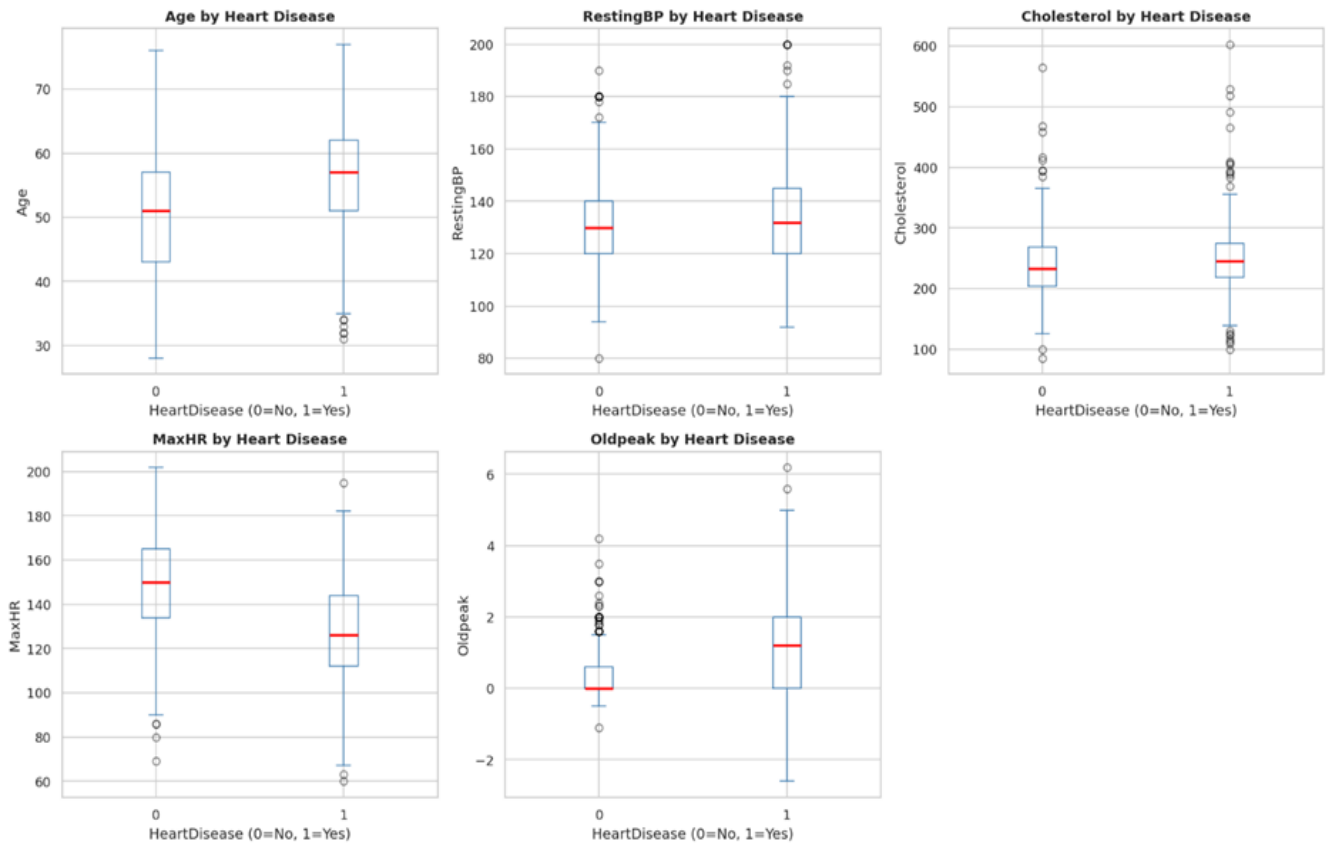


Figure 9: Boxplots of Numeric Features Showing Outliers That are Retained

2.1.4 Multicollinearity (VIF)

VIF scores for all numeric features came in well below the conventional threshold of 5: Age at 1.29, MaxHR at 1.18, RestingBP at 1.10, Oldpeak at 1.09, Cholesterol at 1.01. No features were dropped. Each predictor contributes independent variance to the models, which matters particularly for Logistic Regression and SVM where correlated inputs inflate coefficient instability.

Table 5: Variance Inflation Factor for Numeric Features (Threshold < 5)

Feature	VIF	Interpretation
Age	1.29	No collinearity
RestingBP	1.10	No collinearity
Cholesterol	1.01	No collinearity
MaxHR	1.18	No collinearity
Oldpeak	1.09	No collinearity

2.1.5 Encoding

One-hot encoding with `drop_first` was applied uniformly across all five categorical features. For binary features, Sex and ExerciseAngina, this produces a single column of 0s and 1s, which is equivalent to label encoding but keeps the pipeline consistent. For ChestPainType (4 categories), RestingECG (3), and STSlope (3), `drop_first` removes one redundant dummy column per feature to avoid multicollinearity. This expanded the feature set from 11 to 15 columns.

Table 6: Categorical Feature Encoding Summary (15 Features After Encoding)

Feature	Type	Categories	Encoding
Sex	Binary	M, F	One-hot (drop first)
ChestPainType	Nominal	ATA, NAP, ASY, TA	One-hot (drop first)
RestingECG	Nominal	Normal, ST, LVH	One-hot (drop first)
ExerciseAngina	Binary	Y, N	One-hot (drop first)
ST_Slope	Nominal	Up, Flat, Down	One-hot (drop first)

2.1.6 Scaling

StandardScaler (zero mean, unit variance) was applied to the three distance and gradient-sensitive classifiers: SVM, Logistic Regression, and MLP. The scaler was wrapped inside a scikit-learn Pipeline with the classifier, so it fitted only on each training fold during cross-validation but never on the held-out fold. Tree-based models (Gradient Boosting, Random Forest) and Naive Bayes were left unscaled since their split mechanisms are scale-invariant.

2.2 Classification Models:

Six algorithms were selected to represent a broad spread of modelling strategies: linear, bagging ensemble, boosting ensemble, kernel-based, neural network, and probabilistic. Choosing diverse approaches makes the comparison more informative than running multiple models from the same family, as the results can reveal how much the underlying learning strategy matters for this specific clinical task (Chicco & Jurman, 2020).

GridSearchCV with 5-fold inner cross-validation (scored on F1) handled hyperparameter tuning for all six models. Final performance was measured on 10-fold stratified outer cross-validation, described in Section 2.3.

2.2.1 Gradient Boosting

Gradient Boosting does not need scaling. It splits on value thresholds, so whether cholesterol is in the hundreds or a one-hot column sits at 0 or 1 makes no difference. Table 7 lists the baseline, grid, and tuned parameters.

Table 7: Gradient Boosting Hyperparameter Configuration

Parameter	Baseline	Grid Searched	Tuned
n_estimators	100	100, 200, 300	100
learning_rate	0.1	0.05, 0.1, 0.2	0.05
max_depth	3	3, 4, 5	3
loss	log_loss	-	log_loss

The tuner halved the learning rate from 0.1 to 0.05 but did not touch the tree count or depth. At 0.1, each tree was correcting too aggressively for 917 patients. Dropping to 0.05 forces smaller corrections per tree, which means the ensemble is less likely to latch onto fold-specific noise. The model did not become more powerful; it became more stable. Ahamad et al. (2023) tuned gradient boosting on a comparable heart disease dataset using GridSearchCV and found that lower learning rates with controlled tree counts consistently outperformed aggressive configurations. Here, the count stayed at 100 because the dataset is small enough that convergence happens quickly, even at the slower rate. Max_depth staying at 3 matters because it limits each tree to split on at most three features. That is enough to pick up two-way clinical interactions, say high Oldpeak combined with a flat ST_Slope, without modelling noise. The grid offered depths of 4 and 5, but in boosting each tree fits the residual errors left by the previous trees. A deeper tree can fit those residuals too closely instead of correcting them in a way that holds on to unseen patients. Depth 3 was enough to capture two-way interactions without chasing noise. Bentejac et al. (2021) compared gradient boosting variants across multiple datasets and found that shallow trees with controlled learning rates consistently generalise better than deeper configurations on structured tabular data.

2.2.2 Random Forest

Random Forest does not need scaling. It splits on value thresholds at each node, comparing a feature value against a cut-off to decide which branch a patient follows. Whether cholesterol is in the hundreds or a binary column sits at 0 or 1 has no effect on that decision. Table 8 lists the full configuration.

Table 8: Random Forest Hyperparameter Configuration

Parameter	Baseline	Grid Searched	Tuned
n_estimators	100	100, 200, 300	100
max_depth	None	5, 10, 15	15
min_samples_leaf	1	1, 2, 3	2
Scaling	None	—	None

The two interesting results here are max_depth and min_samples_leaf. The baseline had no depth limit, meaning trees could grow until every leaf was pure. GridSearchCV capped depth at 15 and required at least 2 samples per leaf. Both changes pull in the same direction: they stop individual trees from memorising the training data. An unconstrained tree of 917 patients can grow deep enough to essentially learn specific patients by heart. Capping at 15 and requiring 2 samples per leaf forces each tree to generalise rather than memorise. The ensemble still gets diversity through bootstrap sampling and random feature subsets at each split, so the pruning does not remove the forest's ability to capture non-linear interactions. It removes its tendency to overfit them.

n_estimators staying at 100 reflects the dataset size. More trees would keep averaging, but with 917 patients the variance reduction from going beyond 100 trees is marginal. The added computation does not return proportional gains.

2.2.3 Logistic Regression

Logistic Regression went into a Pipeline with StandardScaler. The model computes a weighted sum of all 15 features and passes it through a logistic function to produce a probability. If cholesterol is sitting at 250 while a one-hot column is at 0 or 1, the raw magnitude difference will dominate the weight updates during optimisation and the model will not learn the smaller-scale features properly. The scaler fits on the training fold only per iteration. Table 9 lists the full configuration.

Table 9: Logistic Regression Hyperparameter Configuration

Parameter	Baseline	Grid Searched	Tuned
C (regularisation)	1.0	0.01, 0.1, 1, 10	0.01

solver	lbfgs	lbfgs, liblinear	lbfgs
max_iter	1000	—	1000
Scaling	StandardScaler in Pipeline	—	StandardScaler in Pipeline

The tuner dropped C from 1.0 all the way to 0.01, which is the most aggressive regularisation option in the grid. C controls the trade-off between fitting the training data tightly and keeping the model coefficients small. A low C means the model accepts more training errors in exchange for simpler, more stable coefficients. On a dataset of 917 patients with 15 features, that trade-off makes sense. The feature set includes correlated clinical variables like Age and MaxHR where unrestricted coefficients would amplify noise rather than signal. The solver staying at lbfgs is consistent with this, as lbfgs handles L2 regularisation efficiently and converges reliably at small C values.

2.2.4 Support Vector Machine

SVM computes distances between patients to draw the class boundary, so it went into a Pipeline with StandardScaler. Without scaling, cholesterol at 250 would overwhelm a one-hot column at 0 or 1 in every distance calculation, regardless of which feature actually separates the classes better. The scaler fits on training data only per fold (Kapoor and Narayanan, 2023). Table 10 shows the full parameter configuration.

Table 10: Support Vector Machine Hyperparameter Configuration

Parameter	Baseline	Grid Searched	Tuned
kernel	rbf	rbf, linear	linear
C	1.0	0.1, 1, 10	1
gamma	scale	scale, auto	scale
Scaling	StandardScaler (in Pipeline)	-	StandardScaler (in Pipeline)

The switch from RBF to linear is the interesting result here. RBF maps data into a higher-dimensional space, looking for curved boundaries. Linear draws a flat hyperplane. GridSearchCV tried both and the flat surface did just as well. For 15 features that are a mix of scaled numeric and binary indicators, the classes apparently separate along straight axes about as cleanly as any curve could manage. RBF's extra flexibility did not find anything linear missed. On a dataset of 917 patients, that is not surprising: a more complex boundary needs more data to justify itself, and here the simpler option generalised just as well. C not moving

from 1 confirms the same story. The default trade-off between fitting the training points tightly and leaving margin for unseen patients was already correct.

2.2.5 Naive Bayes:

Naive Bayes does not need scaling. The model estimates class-conditional Gaussian distributions for each feature independently, then applies Bayes' theorem to compute the posterior probability of each class. The magnitude of a feature has no effect on the probability calculation because the model fits a separate mean and variance for each feature within each class. Extreme values in cholesterol or blood pressure are simply captured in a wider variance estimate for that feature. Table 11 lists the configuration.

Table 11: *Naive Bayes Hyperparameter Configuration*

Parameter	Baseline	Grid Searched	Tuned
var_smoothing	1e-09	1e-9, 1e-8, 1e-7, 1e-6, 1e-5	1e-05
Scaling	None	—	None

The only tunable parameter is var_smoothing, which adds a fraction of the largest observed variance to every feature's class-conditional variance estimate. Its purpose is numerical stability. Without it, features with very small within-class variance produce near-zero denominators in the probability calculation, which causes instability. The selected value of 1e-05 is four orders of magnitude above the default of 1e-09. That large a jump suggests several features in this dataset have genuinely small class-conditional variances, particularly the one-hot encoded binary columns, where within-class variance can be close to zero when one class strongly favours a particular category.

2.2.6 MLP Neural Network

MLP sat in a Pipeline with StandardScaler. Neural networks multiply inputs by weights and sum them at every neuron. When cholesterol sits in the hundreds and binary columns sit at 0 or 1, the weight updates during backpropagation are driven almost entirely by the large-magnitude features. The network ends up ignoring the smaller ones because their gradients are too small to compete.

Table 12: MLP Neural Network Hyperparameter Configuration

Parameter	Baseline	Grid Searched	Tuned
hidden_layer_sizes	(64, 32)	(64,32), (128,64), (64,32,16)	(128, 64)
activation	relu	relu, tanh	relu
alpha (L2)	0.0001	0.0001, 0.001	0.0001
solver	adam	-	adam
learning_rate_init	0.001	-	0.001
max_iter	500	-	500
early_stopping	True (10% val)	-	True (10% val)
Scaling	StandardScaler (in Pipeline)	-	StandardScaler (in Pipeline)

The tuned models doubled the layer widths from (64, 32) to (128, 64), although it dismissed the three-layer option (64, 32, 16). After encoding with 15 features now, 64 neurons in the first layer were not enough to represent useful combinations. Doubling that combination to 128 gave the network more room. A third layer would have added another level of abstraction that 917 patients cannot support without memorising training examples. Alpha remained at [0.0001], the lighter of the two regularisation options, because early stopping and data leakage were already handled by overfitting. Training stops automatically when validation loss plateaus (early stopping), so the lighter regularisation at alpha 0.0001 was sufficient. Adding heavier weight decay on top would have restricted the network more than necessary. Accuracy came to 0.8549, F1 0.8723, AUC 0.9144. Last of the six. The other five models all scored within 0.75 F1 points of each other, and MLP sits 0.8 points below that cluster. Its confusion matrix has more errors on both sides along with more heart disease patients missed and more healthy patients wrongly flagged.

2.3 Evaluation framework

2.3.1 Cross-validation design

All six classifiers were evaluated using 10-fold stratified cross-validation. Stratified means each fold keeps the same 55.3% to 44.7% class split as the full dataset, so no fold ends up with a skewed ratio. Every patient appears in the validation set exactly once across the 10 iterations. Reported metrics are averages over all folds.

There are two layers. The outer loop is the 10-fold stratified CV that produces the final performance numbers. Inside each outer training fold, GridSearchCV runs a separate 5-fold CV to pick hyperparameters. Tuning and evaluation never share data. If they did, the scores would reflect how well the model fits its tuning set, not how it handles unseen patients. Prusty et al. (2022) used a similar stratified setup for disease classification and noted that stratification matters most below 1,000 observations, since random splits at that size can easily produce folds with uneven class distributions.

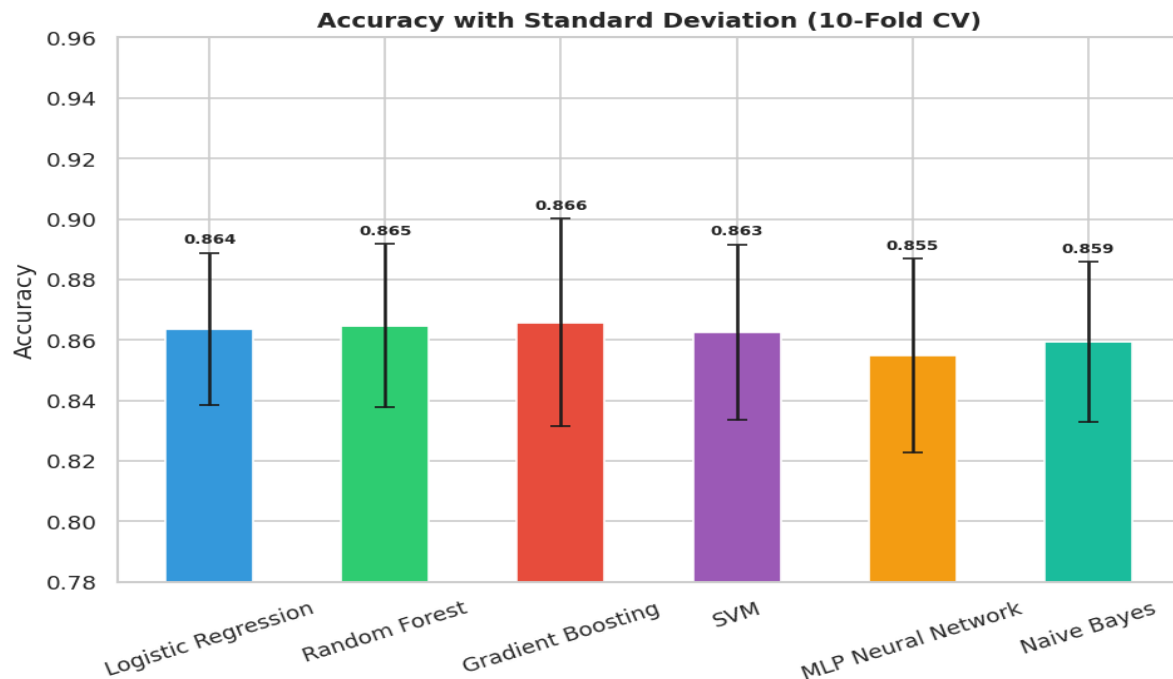


Figure 10: 10-Fold Stratified Cross-Validation Accuracy with Standard Deviation for All Six Classifiers

2.3.2 Pipeline design

SVM, Logistic Regression, and MLP only these models are placed inside a scikit-learn Pipeline with StandardScaler as the first step. When the cross-validation step takes place, the scaler is fitted on the training portion of each fold and transforms the validation portion of the fitted values from the scaler. The alternative, scaling the entire dataset before splitting, leaks validation information into the scaling step. Kapoor and Narayanan (2023) reviewed 329 ML papers and found this type of preprocessing leakage in many studies, often inflating reported accuracy by several points.

Gradient Boosting, Random Forest, and Naive Bayes did not need Pipelines. They split on value thresholds or compute probabilities, and neither operation depends on feature scale.

2.3.3 Scoring metrics

Three metrics were recorded for each model:

Table 13: Scoring Metrics Used

Metric	What it measures	Why it matters here
Accuracy	Proportion of correct predictions overall	Baseline, but limited: predicting heart disease for everyone scores 55.3%
F1 score	Harmonic mean of precision and recall	Balances missed patients (low recall) against false alarms (low precision). Used as GridSearchCV criterion
AUC	Ranks patients by risk across all thresholds	Broader picture when models are close on accuracy and F1. 0.5 = random, 1.0 = perfect

2.3.4 Statistical comparison methodology and the Statistical Tests

Two tests were used to check whether performance differences between models are significant or just cross-validation data leakage.

Paired t-test: Two classifier accuracy scores were compared. Considering both models train and test on the same 10 iterations, the comparison is considered paired, and a paired t-test can be done. The difference in p-value under 0.05 is unlikely to be random. Raschka (2020) recommended this paired approach for single-dataset comparisons because unpaired tests ignore the shared fold structure and tend to make differences look more significant than they actually are.

McNemar's test: Works at the patient level. For each pair of classifiers, it counts patients where one model is right and the other is wrong, then checks if that disagreement goes mostly one direction. If it does, the two classifiers make systematically different errors.

3. Results

3.1 Classification Performance

Figure 11 shows the tuned performance for all six classifiers side by side. Table 14 ranks them by F1 after tuning. Only 0.75 F1 points separate first place from fifth, and MLP is the only model that sits visibly below the group.

Table 15 consolidates the per-class classification reports for all six models. The precision, recall, and F1 figures shown here are discussed individually in each model's subsection below, alongside the hyperparameter configurations and confusion matrices that produced them. The table is placed here so the reader can compare across models without flipping between subsections.

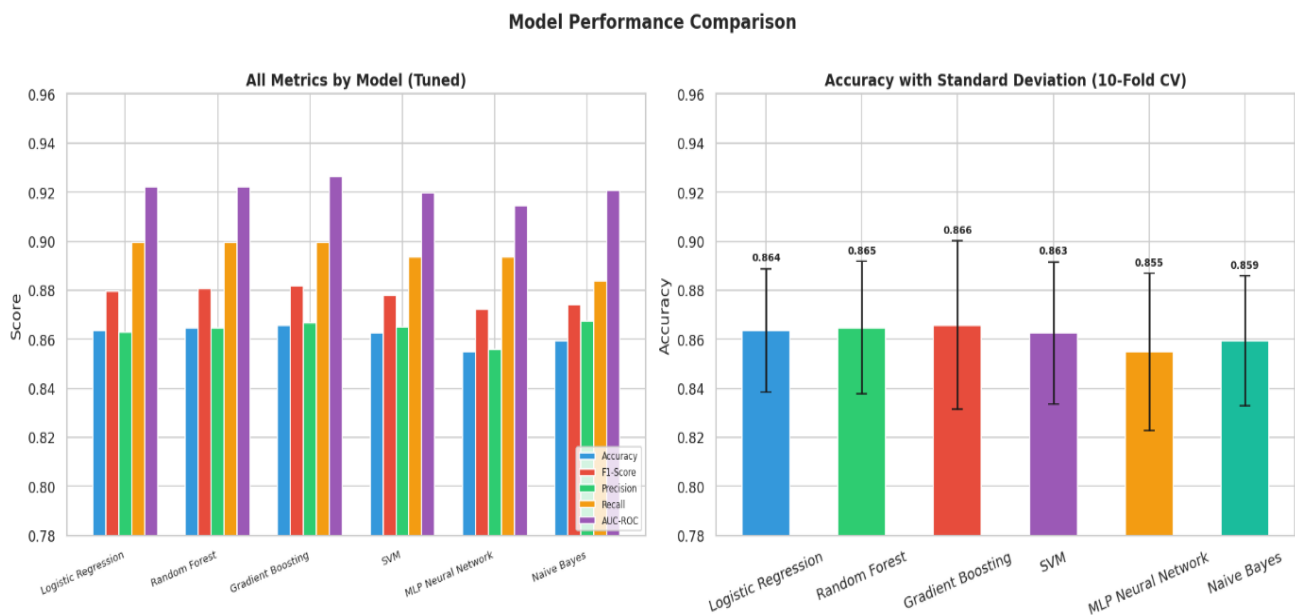


Figure 11: Tuned Performance Summary for All Six Classifiers

Table 14: Model Rankings by F1 Score After Tuning

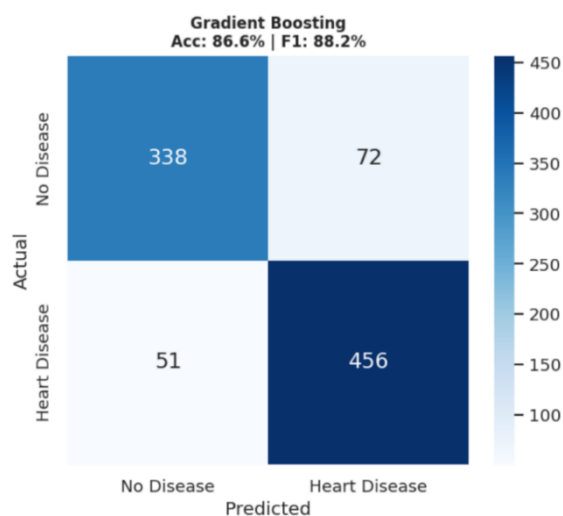
Rank	Model	Accuracy	F1	AUC
1	Gradient Boosting	0.8658	0.8815	0.9262
2	Random Forest	0.8647	0.8806	0.9220
3	Logistic Regression	0.8636	0.8795	0.9222
4	SVM	0.8626	0.8779	0.9195
5	Naive Bayes	0.8593	0.8740	0.9209
6	MLP Neural Network	0.8549	0.8723	0.9144

Table 15: Per-Class Precision, Recall, And F1 for All Six Tuned Classifiers (10-Fold CV On 917 Patients)

Model	Class	Precision	Recall	F1	Support	Accuracy
Gradient Boosting	No Disease	0.87	0.82	0.85	410	0.87
	Heart Disease	0.86	0.90	0.88	507	
Random Forest	No Disease	0.87	0.82	0.84	410	0.86
	Heart Disease	0.86	0.90	0.88	507	
Logistic Regression	No Disease	0.87	0.82	0.84	410	0.86
	Heart Disease	0.86	0.90	0.88	507	
SVM	No Disease	0.86	0.82	0.84	410	0.86
	Heart Disease	0.86	0.89	0.88	507	
Naive Bayes	No Disease	0.85	0.83	0.84	410	0.86
	Heart Disease	0.86	0.88	0.87	507	
MLP Neural Network	No Disease	0.86	0.81	0.83	410	0.85
	Heart Disease	0.85	0.89	0.87	507	

3.1.1 Gradient Boosting

Tuned accuracy results were [0.8658, F1 0.8815, AUC 0.9262], placing it first overall in the 6 models tested. Baseline had been 0.8592 accuracy and 0.8751 F1, so tuning the models increased 0.7 points on accuracy and 0.6 on F1. The default base models were already a



reasonable fit for this data, which explains the small gain. The confusion matrix leans toward false negatives, with more heart disease patients misclassified as healthy than healthy patients misclassified as sick. Of all six models, Gradient Boosting had the fewest total misclassifications. This trend was followed from Grinsztajn et al. (2022), who observed the same across 45 tabular datasets at NeurIPS, and Theerthagiri (2022) confirmed it specifically for cardiovascular data.

Figure 12: Gradient Boosting Confusion Matrix

3.1.2 Random Forest

Tuned accuracy: 0.8647, F1: 0.8806, AUC: 0.9220. Second overall. Baseline had been 0.8570 and 0.8739, so the gain was 0.8 points on accuracy and 0.7 on F1, the second largest improvement of any model after MLP. The training-to-CV gap was 8.62 points, the only model classified as a mild overfit. That gap exists because individual trees still memorise portions of the training data even with the depth and leaf constraints applied. Bagging reduces this substantially but does not eliminate it at this sample size. The gap is worth monitoring but does not undermine the model's practical usefulness. Recall for Heart Disease was 0.90, matching Logistic Regression and Gradient Boosting exactly. Barreñada et al. (2024) noted that random forest overfitting on probability estimation is a known behaviour on moderate-sized clinical datasets, and that CV performance remains a reliable indicator of real-world generalisation even when the training gap looks elevated.

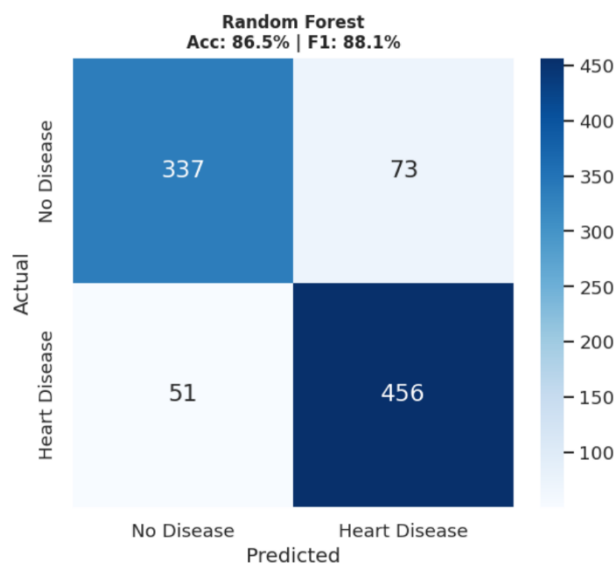
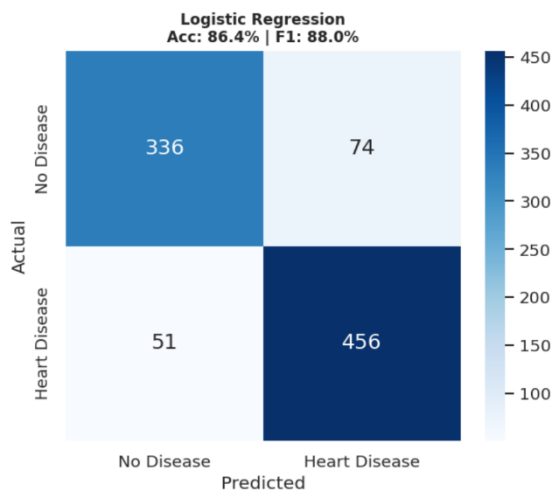


Figure 13: *Random Forest Confusion Matrix*

The confusion matrix shows a similar error distribution to the other top models, with false negatives slightly outnumbering false positives. The feature importance scores extracted from this model placed STSlopeUp, Oldpeak, and ExerciseAngina at the top, consistent with the clinical literature and the Point-Biserial findings from the EDA.

3.1.3 Logistic Regression

Tuned accuracy: 0.8636, F1: 0.8795, AUC: 0.9222 coming third best overall. Baseline had been 0.8615 and 0.8768, so tuning added 0.2 points on accuracy and 0.3 on F1. The gain is modest because the default configuration was already a reasonable fit for this data. What stands out is



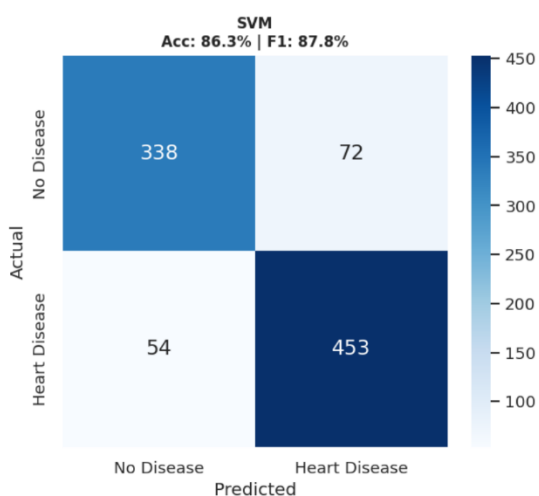
the AUC of 0.9222. For a linear classifier drawing a single flat decision boundary across 15 features, that is a meaningful result. It tells you the two classes separate well along straight axes in this feature space, and non-linear complexity does not add much. The training-to-CV accuracy gap was 0.22 points, the second smallest of all models, confirming that strong regularisation produced a highly stable and generalisable classifier.

Figure 14: *Logistic Regression Confusion Matrix*

The confusion matrix shows the model missing more heart disease patients than it incorrectly flags healthy ones, which is the same directional pattern as Gradient Boosting and SVM. Recall for the heart disease class was 0.90. Nusinovici et al. (2020) found logistic regression matched or outperformed more complex classifiers on several chronic disease datasets precisely because the underlying relationships were largely linear, which appears to be the case here as well.

3.1.4 Support Vector Machine

Tuned accuracy: 0.8626, F1 0.8779, AUC 0.9195. Fourth overall. Baseline was 0.8603 / 0.8782. Tuning added 0.2 points on accuracy and lost



0.03 on F1. Practically the same model. The training-to-CV gap was 0.44 points, the second smallest of all six, so SVM is among the least overfit classifiers here. The confusion matrix has more false negatives than false positives, similar to Gradient Boosting but with a few more errors overall. Alsabhan and Alfadhly (2025) placed SVM in a comparable position on a similar heart disease dataset.

Figure 15: *SVM Confusion Matrix*

3.1.5 Naive Bayes

Tuned accuracy: 0.8593, F1: 0.8740, AUC: 0.9209. Fifth on accuracy and F1, but fourth on AUC. That ordering is informative. At the fixed 0.5 decision threshold, Naive Bayes makes slightly more combined errors than SVM. But across all thresholds, its probability rankings are better than SVM's. The model is producing well-calibrated posterior probabilities even though its feature independence assumption is technically violated. The clinical features here are mildly correlated, as confirmed by the VIF and heatmap analyses, but not enough to substantially distort the probability estimates. Chicco and Jurman (2020) observed the same behaviour on the heart failure dataset, where Naive Bayes consistently produced competitive AUC values despite being the simplest model in the comparison.

The training-to-CV gap was 0.11 points, the smallest of all six models. Naive Bayes has almost no capacity to overfit. Its model consists entirely of per-class means and variances, and there are no complex interactions to memorise.

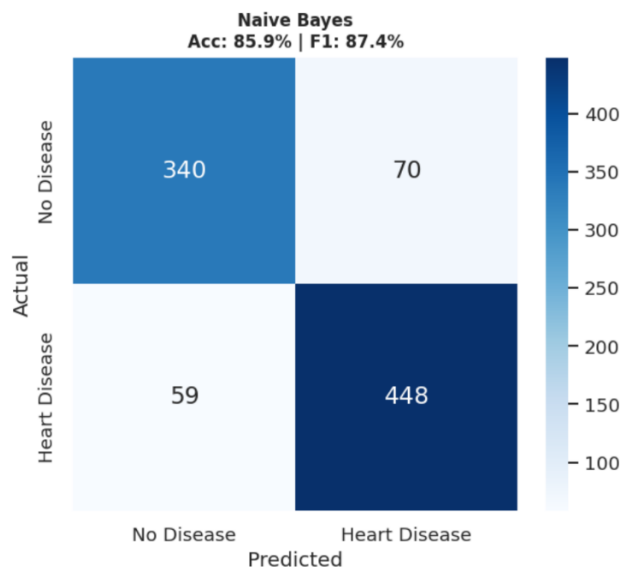


Figure 16: Naive Bayes Confusion Matrix

The confusion matrix shows a slightly more balanced error distribution than the top three models, with recall for HeartDisease at 0.88 compared to 0.90 for the others. Fewer true disease patients are caught, but the false alarm rate on healthy patients is also slightly lower.

3.1.6 MLP Neural Network

The underperformance is structural, not a tuning failure. Tree-based models split recursively on individual feature thresholds, which naturally produces axis-aligned, irregular decision

boundaries. That is the kind of boundary clinical tabular data tends to need, where a patient might be at high risk because Oldpeak is above 1.5 AND ST_Slope is flat, but not because of some smooth continuous function across all 15 features. Neural networks learn smooth boundaries, and they need thousands of examples to approximate the irregular shapes that trees capture with a few hundred splits. Grinsztajn et al. (2022) tested this across 45 datasets at NeurIPS and found trees winning consistently below several thousand samples. Borisov et al. (2024) reported the same for cardiovascular data at this scale.

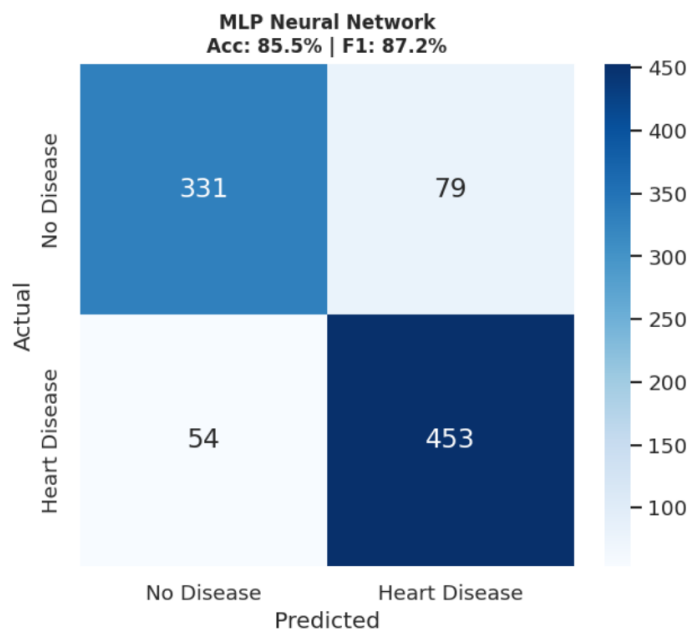


Figure 17: MLP Confusion Matrix

3.2 Scoring Interpretation

Accuracy alone cannot separate the models here. The top five classifiers fall within about 1 percentage point of each other. F1 compresses both types of error into one number: low precision means healthy patients flagged, low recall means sick patients missed. AUC captures ranking ability across every possible decision threshold, not just the default 0.5.

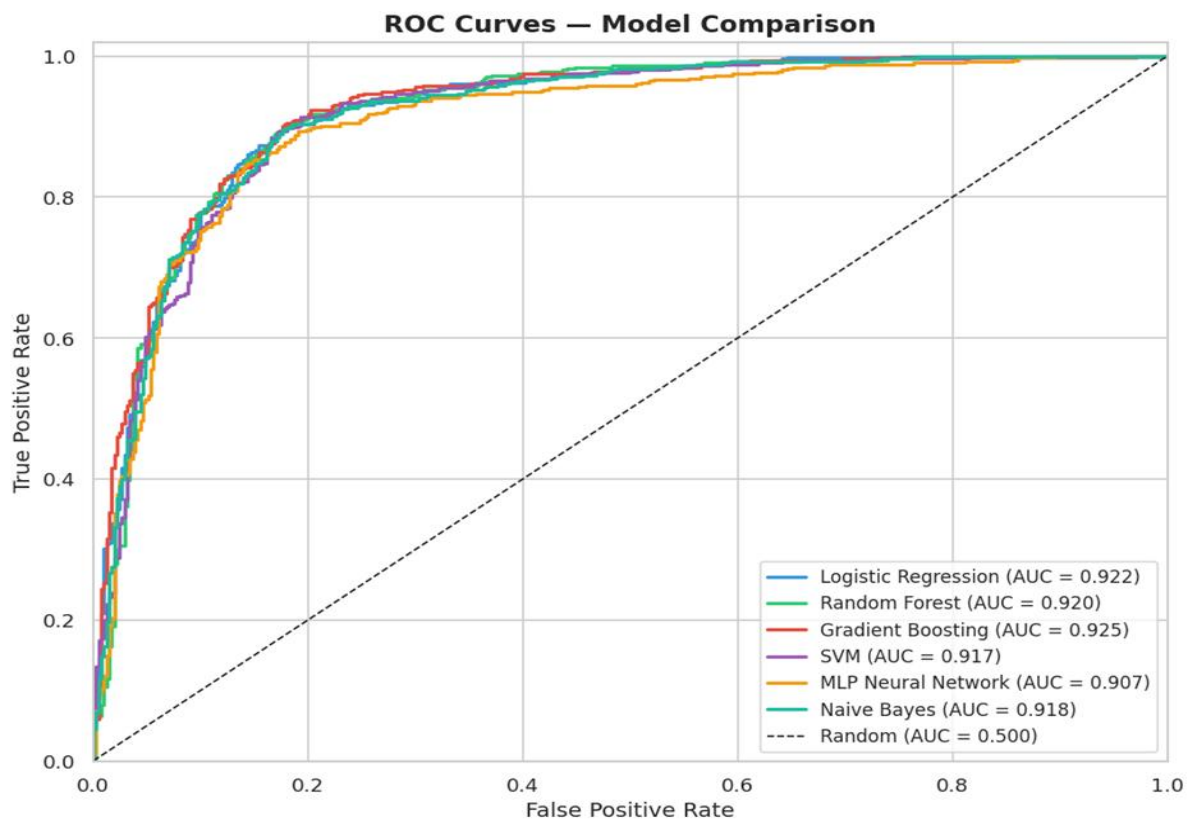


Figure 18: ROC Curves for All Six Classifiers Overlaid

3.3 Statistical Test Results

Table 16: Paired T-Test Results (Selected Comparisons)

Model A	Model B	t-stat	p-value	Significant
Gradient Boosting	SVM	0.6048	0.5602	No
Gradient Boosting	MLP Neural Network	1.1387	0.2842	No
Gradient Boosting	Naive Bayes	0.9705	0.3572	No
Random Forest	MLP Neural Network	0.9746	0.3552	No
SVM	MLP Neural Network	0.9610	0.3617	No
Logistic Regression	MLP Neural Network	1.2411	0.2459	No

No pairwise comparison produced a p-value below 0.05. The top five classifiers sit within about 1 accuracy point of each other. Even MLP, the furthest behind, trails by only about 1 point on accuracy. With 10 folds of roughly 92 patients each, the fold-to-fold variance is large enough that the test cannot separate between-model differences from noise. The largest t-statistic was

Logistic Regression vs MLP at 1.24 ($p = 0.25$). On a bigger dataset some of these comparisons would probably reach significance, particularly Gradient Boosting vs MLP. On 917 patients the differences are real in direction but not confirmable at $\alpha = 0.05$.

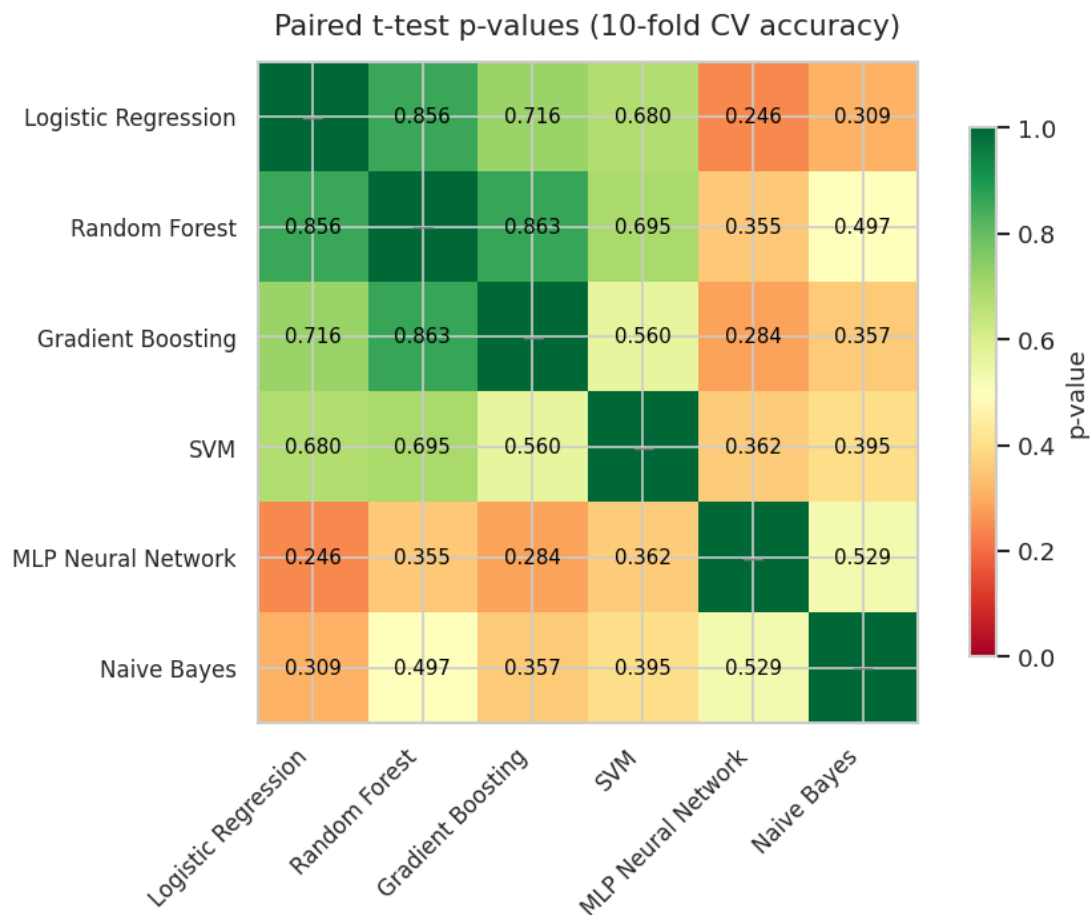


Figure 19: Paired T-Test P-Values for Pairwise Accuracy Comparison Across 10-Fold CV

Table 17: McNemar's Test Results (Selected Comparisons)

Model A	Model B	Chi-sq	p-value	Significant?
Gradient Boosting	SVM	0.0635	0.8011	No
Gradient Boosting	MLP Neural Network	1.1571	0.2821	No
Logistic Regression	MLP Neural Network	1.5312	0.2159	No
Random Forest	MLP Neural Network	1.1228	0.2893	No
Logistic Regression	Random Forest	0.0000	1.0000	No
SVM	Naive Bayes	0.1143	0.7353	No

None reached $p < 0.05$. The classifiers are not only scoring similarly; they are mostly agreeing on the same patients. Where they disagree, neither one is consistently right while the other is wrong. The two tests agree. Six classifiers on 917 patients with 15 features all land in a narrow band. Gradient Boosting tops the ranking and MLP trails, but the gaps are too small to confirm statistically on this sample size.

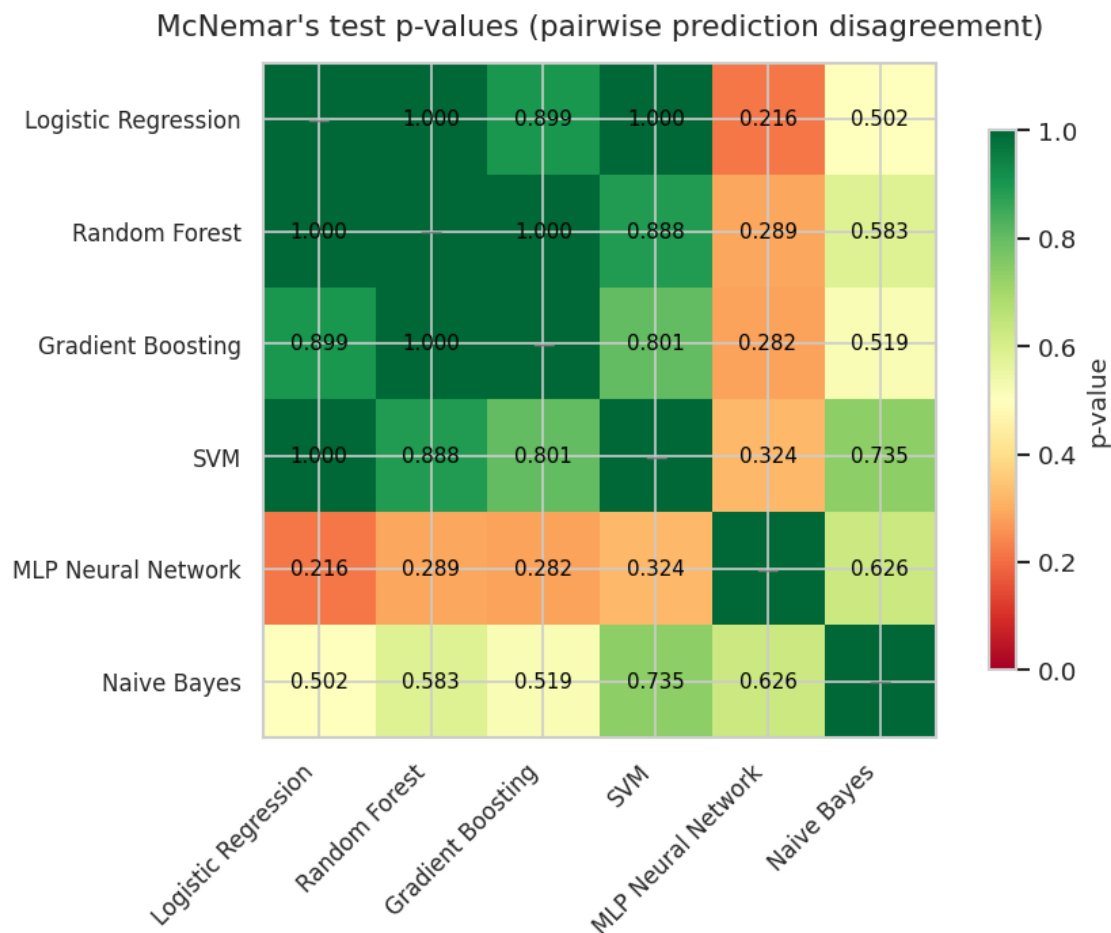


Figure 20: McNemar's Test P-Values for Pairwise Prediction Disagreement

3.4 Feature Importance

Feature importance was extracted from three models, i.e Gradient Boosting, Random Forest, and Logistic Regression. It is normalised against each model's top feature to enable direct comparison across Table 18.

Gradient Boosting ranks features by the cumulative reduction in log-loss across all trees. STSlopeUp dominated with a normalised score of 1.000, while all other features scored below 0.20. This sharp concentration reflects how boosting works. Early trees lock onto the single strongest split, and subsequent trees correct residuals on increasingly narrow subsets of

patients. Oldpeak (0.183) and MaxHR (0.134) were the next most informative, both capturing cardiac response under physical stress.

Feature importance is spread out more evenly among features when using a Random Forest model than with other models. Again, the STSlopeUp feature ranked first in terms of importance (1.000), followed by STSlopeFlat (0.760), Oldpeak (0.658), ExerciseAngina (0.560), and MaxHR (0.543) which also have significant relative importance. The more broadly distributed feature importance is indicative of the independence of the ensemble; since each tree in the ensemble runs separately and chooses features randomly at each splitting of the data, feature importance accumulates over many different combinations of features instead of concentrating on one single dominating feature.

Logistic Regression importance is derived from the absolute values of the standardised coefficients. STSlopeUp ranked first (1.000), followed by ChestPainTypeATA (0.851), ExerciseAngina (0.797), and SexM (0.770). The higher relative weight of categorical features here reflects how logistic regression processes one-hot encoded variables. Each binary column receives an independent coefficient, and categories with strong class separation naturally attract larger absolute values after scaling.

Table 18: Normalised Feature Importance Across Three Models

Feature	GB	RF	LR	Average
ST_Slope_Up	1.000	1.000	1.000	1.000
ST_Slope_Flat	0.024	0.760	0.890	0.558
Oldpeak	0.183	0.658	0.627	0.489
ExerciseAngina_Y	0.109	0.560	0.797	0.489
MaxHR	0.134	0.543	0.516	0.398
ChestPainType_ATA	0.064	0.240	0.851	0.385
Sex_M	0.118	0.238	0.770	0.375
ChestPainType_NAP	0.092	0.150	0.705	0.316
FastingBS_1.0	0.072	0.181	0.680	0.311
Age	0.050	0.400	0.316	0.255

ST_Slope_Up came out on top across all three models. A flat or upsloping ST segment during exercise is a known clinical sign of ischaemic heart disease, so the model is picking up real

physiology, not noise. Oldpeak, which measures ST depression from exercise relative to rest, also ranked high. Both features capture how the heart responds under stress. The models weighted them well above resting measurements like blood pressure or cholesterol.

Cholesterol ranked low despite being the feature most people associate with heart disease. It had the lowest VIF (1.01) and after imputation its distribution did not separate the two classes as cleanly as the exercise variables did. Teja and Rayalu (2025) reported a similar ordering in their BMC study, with exercise variables dominating. Additionally, Alsabhan and Alfadhly (2025) found ST_Slope and Oldpeak among the top predictors on a comparable dataset.

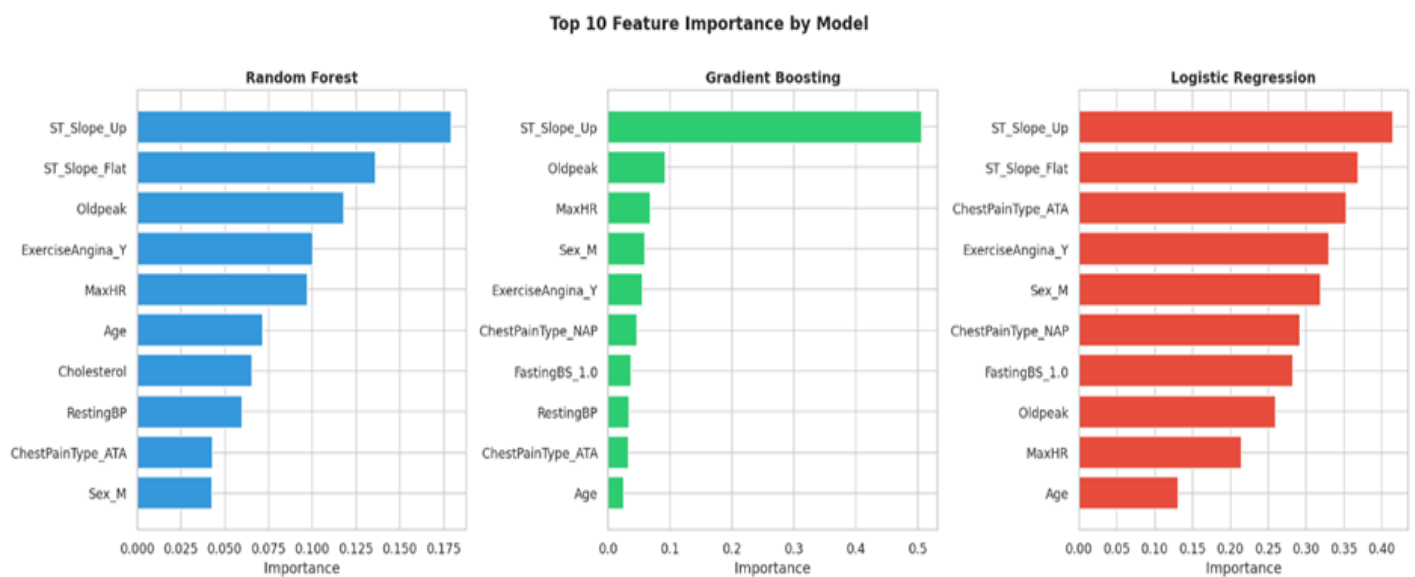


Figure 21: Feature Importance Bar Chart for the Top 3 Models

3.5 Learning Curves

Learning curves were generated for all six tuned models using training set sizes from 10% to 100% of the data (10 steps), with 10-fold stratified CV at each size.



Figure 22: Learning Curves for All Six Classifiers (2x3 Grid)

What the curves show for each model:

Gradient Boosting: Training accuracy starts high and stays above validation throughout, but the two lines converge as samples increase. By 917 patients they are about 4.8 points apart. The validation curve is still climbing slightly at the right edge, which means more data would likely improve performance further.

Random Forest: For all sizes of training, training accuracy holds steadily at 0.95-0.96; yet, validation curve grows steadily from 0.80 to 0.86, resulting in an 8.62-point difference between curves upon completion. The cross-validation line is continuing to increase at the far right of the graph, indicating the model has not yet reached its ceiling. Random forest has a higher capacity than the current data set can utilise; if more labels were available, it would further reduce this gap. Barreñada et al. (2024) noted this same phenomenon in their analysis of clinical datasets of similar size.

Logistic Regression: Training accuracy starts at 0.84, peaks around 0.93 at 300 samples, then drops back toward 0.86 as the full dataset regularises the coefficients. The validation curve rises steadily from below and the two lines converge tightly by 825 patients, ending just 0.22 points apart. The convergence from opposite directions confirms a well-generalising model. The plateau from around 600 samples onward indicates the feature set has been fully utilised under a linear boundary.

SVM: Training and validation curves almost overlap. By 917 patients the gap is just 0.44 points. There is nothing left for the model to gain from more data or a more complex kernel. It has learned what the features it can output and stopped there.

Naive Bayes: Training accuracy begins around 0.90, rises slightly, then gradually drops toward 0.86 as more data stabilises the variance estimates. The validation curve climbs from 0.80 and the two lines converge by around 600-700 samples, ending just 0.11 points apart at full size. The convergence is steady rather than immediate, reflecting that the per-class Gaussian estimates need a reasonable number of patients to stabilise. Once they do, the model neither improves nor degrades with additional data, confirming the ceiling is set by the independence assumption rather than sample size.

MLP Neural Network: Training and validation curves are closer together than expected for a neural network (gap of 1.54 points at full size), but validation accuracy is lower than every other model at every training size. The network is not overfitting badly because early stopping prevents that. It is just not extracting as much signal from the features as the other classifiers.

The overall pattern is convergence. No validation curve is dropping at full training size, so none of the models are dramatically overfitting. But none are climbing steeply either. The statistical tests in Section 3.3 back this up: with 917 patients and 15 features, the data has a performance ceiling that does not move much regardless of which algorithm sits on top.

3.6 Overfitting Analysis

Table 19 compares training accuracy (fitted on all 917 samples) against mean cross-validation accuracy. A gap below 5% was classified as Good, 5-10% as Mild Overfit, and above 10% as Overfitting.

Table 19: Training Vs Cross-Validation Accuracy Gap for All Tuned Models

Model	Train Acc (%)	CV Acc (%)	Gap (%)	Status
Logistic Regression	86.59	86.36	0.22	Good
Random Forest	95.09	86.47	8.62	Mild Overfit
Gradient Boosting	91.38	86.58	4.80	Good
SVM	86.70	86.26	0.44	Good
MLP Neural Network	87.02	85.49	1.54	Good
Naive Bayes	86.04	85.93	0.11	Good

Five out of six models exhibit nearly identical accuracies across their train and cross-validation datasets, with only slight differences (0.22% and 0.11% for logistic regression and naive bayes respectively). Both logistic regression and naive bayes have the strictest structural assumptions (linearity and independence of features), preventing them from learning very specific sequential patterns in their training sets.

Random Forest is the only model classified as a mild overfit, with a 8.62% gap. Individual trees in a random forest can grow deep enough to memorise subsets of the training data, and while bagging reduces this substantially, it does not eliminate it on a dataset of 917 samples (Barreñada et al., 2024). The CV metrics remain strong (accuracy 0.8647, F1 0.8806), so this overfitting does not impair practical utility. However, deploying this model on a patient population that differs significantly from the training distribution would carry more risk than deploying Logistic Regression under the same conditions.

Gradient Boosting shows a 4.80% gap, sitting just below the mild overfit threshold. Its sequential tree-building process is inherently more aggressive at fitting training data than bagging. The conservative configuration selected by GridSearchCV (`max_depth = 3`, `learning_rate = 0.05`, `n_estimators = 100`) is what keeps this in check.

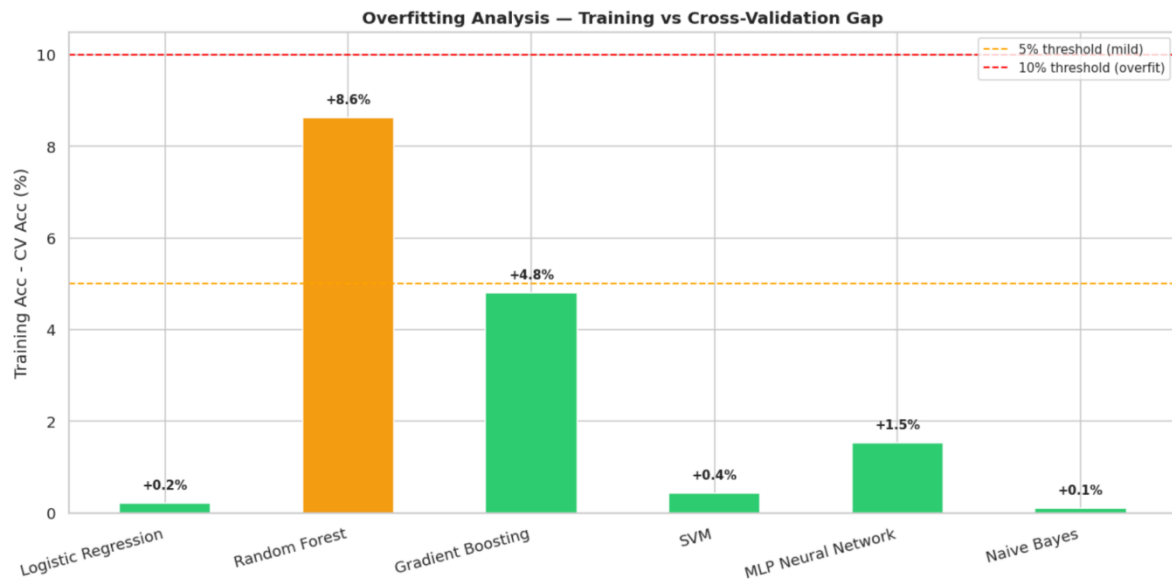


Figure 23: Overfitting Analysis (Training Vs Cross-Validation Accuracy Gap)

3.7 Model Comparison and Clinical Relevance

GB leads on accuracy (0.8658), F1 (0.8815), and AUC (0.9262). However, all six classifiers sit within 1.1 accuracy points and 0.9 F1 points of each other. The paired t-tests confirmed that none of these gaps reached statistical significance for p values below 0.05 on the tested patients. GB wins the ranking, but the margin over RF (0.0009 on F1) and LR (0.0020 on F1) is too narrow to state a conclusive winner. Cardiovascular disease was the sole reason for the death of 20.5 million people in the year 2021, roughly a third of all deaths worldwide (World Health Organization, 2024), and most of those deaths took place in regions where specialist access or Diagnostic imaging machines like echocardiograms, CT angiography, cardiac MRI.. The classifiers here only use measurements that a GP already collects in a routine checkup. If a model at this accuracy level were deployed in those settings, it could catch patients who would otherwise not be referred until after myocardial damage has already occurred.

Recall ranged from 0.8838 (NB) to 0.8996 (LR). In clinical terms, recall is the proportion of actual heart disease patients the model correctly identifies. At 0.90 recall, about 10 in every 100 heart disease patients walk away undetected. They do not get referred, do not get put on statins or beta-blockers, and some present again only after a cardiac event. Alsabhan and Alfadhly (2025) tested a comparable set of classifiers on the same dataset in Scientific Reports and emphasised that sensitivity to disease cases is the most important metric for screening applications, because the cost of a missed diagnosis (delayed treatment, potential cardiac event)

far outweighs the cost of a false alarm (an extra blood test or referral). All six models achieved recall above 0.88, so fewer than 12% of heart disease patients were missed on average. Tiwari et al. (2022) similarly found that ensemble-based frameworks for cardiovascular prediction achieved consistent recall above 0.87 across multiple dataset configurations, reinforcing that well-tuned classifiers on clinical tabular data tend to converge toward a performance ceiling rather than diverge.

Precision works in the opposite direction. NB ranked fifth on F1 but had the highest precision (0.8674) because it was more conservative with positive predictions. Fewer patients were labelled for heart disease, but the ones that did were more likely to actually have cardiological issue. This is important in practice because every positive prediction requires the patient to follow up with their doctor for counselling: an echocardiogram, a stress test, a cardiologist referral. Healthy patients who go through that process unnecessarily face costs, anxiety, and in some cases, invasive procedures that carry their own risk. MLP had the lowest precision at 0.8560. Put differently, about 1 in 7 of its positive predictions landed on a healthy patient.

AUC works at a different level. Rather than looking at a single threshold, it measures whether the model consistently ranks sick patients above healthy ones across all possible cutoffs. GB had an AUC of 0.9262, meaning it assigned the higher risk score to the sick patient correctly 93% of the time when comparing one sick and one healthy patient at random iteration. Moreover, LR came third on accuracy but second on AUC (0.9222), which tells us its probability outputs are well calibrated. Therefore, a doctor who wants to set their own threshold, 0.3 for aggressive screening or 0.7 for conservative, would get reliable probability estimates from LR without retraining. Teja and Rayalu (2025) reported a similar AUC for gradient boosting (0.92) on the same combined heart disease dataset and found that ensemble methods consistently outperformed single-model approaches on this type of clinical tabular data.

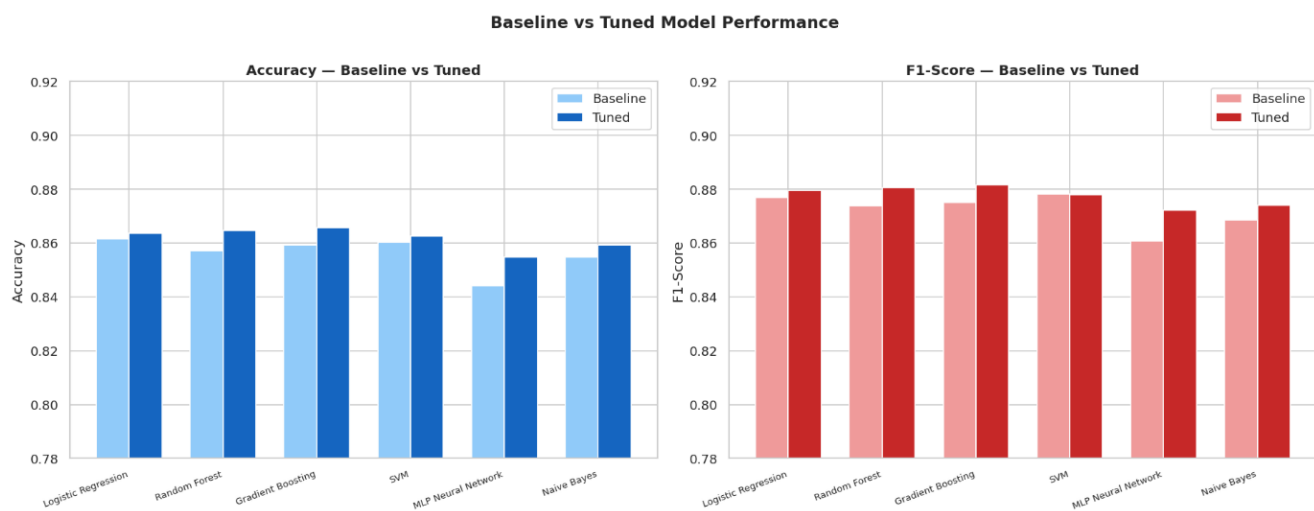


Figure 24: Performance Comparison Bar Chart

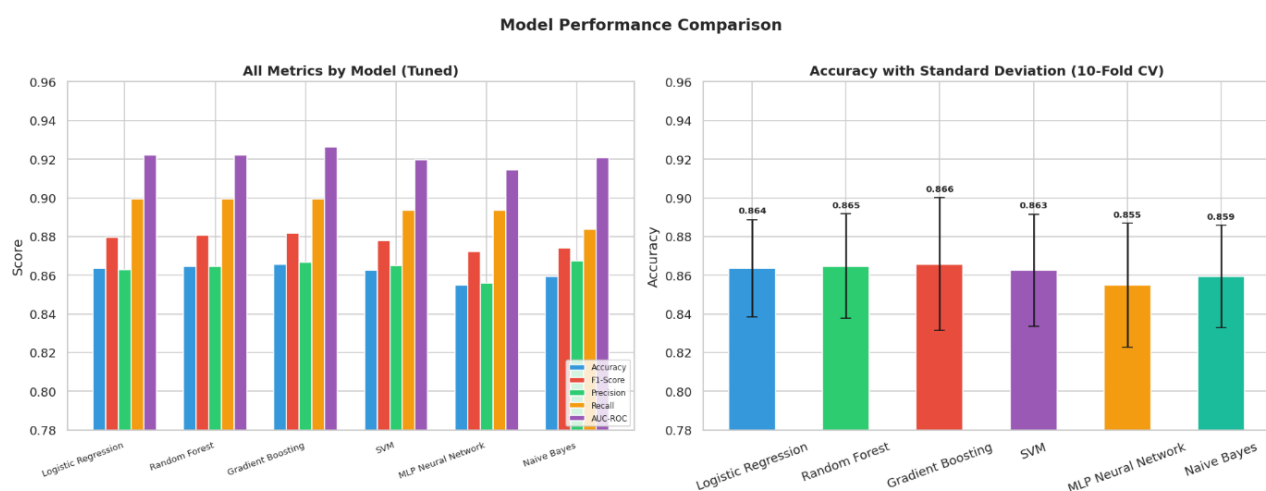


Figure 25: Tuned Model Accuracy with Standard Deviation Across 10-Fold Cross-Validation

4. Conclusion

4.1 Summary of Findings

Six classification models were built and evaluated for binary heart disease prediction on 917 patient records from five cardiovascular databases. The pipeline covered data cleaning, KNN imputation of missing cholesterol values, one-hot encoding, per-fold scaling through sklearn Pipelines, GridSearchCV tuning, and 10-fold stratified cross-validation. Gradient Boosting ranked first on all three metrics, but only 1.09 accuracy points separated it from fifth place. Paired t-tests and McNemar's tests confirmed none of these gaps reached significance at $p < 0.05$. Six fundamentally different algorithms landing in such a narrow band suggests the ceiling

here is set by the dataset, not by the classifier. The clinical features carry consistent signal, and any reasonably configured model extracts most of it (Uddin et al., 2019).

Since the differences are not statistically separable, clinical and operational criteria should drive model selection. LR is the strongest candidate for deployment. Clinics need to trace a prediction back to specific patient features, and regulatory frameworks increasingly require that algorithmic decisions in medicine be explainable (Xu et al., 2023).

4.2 Limitations

The dataset merges records from five cardiovascular databases. All were collected decades ago. Clinical practice, diagnostic thresholds, and population health profiles have changed since then. Statin prescriptions are now routine, cardiac imaging has improved, and lifestyle risk factors look different. A model trained on these records may be fitting patterns that no longer apply to a modern patient walking into a GP clinic today.

Of the 918 records, 725 are male and 193 females. The models have seen far fewer female heart disease records during training. Heart disease in women often looks different: atypical chest pain, more frequent nausea, lower ST segment changes (Liu et al., 2025). The aggregate metrics may overstate how well these models would perform on female patients. The sample is too small for a reliable subgroup analysis to test that directly.

Eleven clinical measurements from a single visit cannot capture the full complexity of cardiovascular risk. How blood pressure changes over repeated visits, how cholesterol responds to medication, whether the patient smokes or exercises: none of this is in the dataset. Genetic markers are also absent. Javeed et al. (2022) observed that models trained on structured data features alone tend to plateau at similar accuracy levels because these features only capture part of the clinical picture. All six classifiers converging in this report is consistent with that.

All reported metrics come from 10-fold cross-validation on the same 917 patients. No external validation was performed on a separate cohort from a different hospital, country, or time period. Liu et al. (2025) stated ML-based cardiovascular prediction lacked external validation. The scores reported here should be treated as optimistic upper bounds. Navarro et al. (2022) found a similar gap, clinical prediction models built with supervised ML frequently omitted calibration and external validation, making real-world reliability difficult to judge.

4.3 Recommendations for Future Work

None of these models should be considered for clinical screening without external validation first. They would need to be tested on independent patient cohorts from different hospitals and demographic populations to show whether the learned patterns actually transfer. Liu et al. (2025) documented consistent performance drops in cardiovascular ML models that underwent external validation. Until that step is done, the numbers in this report describe performance on a benchmark, not on real patients.

The feature space is narrow and needs to be expanded. This dataset captures a single visit in 11 features. Longitudinal EHR data, where patients have multiple visits over time, would let models pick up on temporal trends that a single dataset cannot. A patient whose resting blood pressure has risen three years is clinically different from one whose reading has been stable now. Echocardiogram measurements, coronary calcium scores and patient-reported variables (smoking status, physical activity) would bring in risk factors that the current dataset does not cover. Tiwari et al. (2022) showed measurable gains in cardiovascular prediction when ensemble frameworks were given access to more features.

All six models used a default 0.5 decision threshold. In clinical practice, that number is arbitrary. A primary care clinic running preventive screening might prefer 0.3 to catch more at-risk patients, accepting more false positives. A specialist referral pathway might use 0.7 to reduce unnecessary invasive follow-ups. Future work should evaluate calibration curves and Brier scores for these models to determine how reliably their probability outputs map to actual disease prevalence. Platt scaling or isotonic regression may improve that mapping if raw probabilities turn out to be poorly calibrated.

The gender imbalance (79% male) also needs to be addressed in future studies. Aggregate metrics can hide poor performance on underrepresented groups. Precision, recall, and calibration should be reported separately by sex and age group, and by ethnicity where sample size allows. Alsabhan and Alfadhly (2025) noted that heart disease prediction models trained on skewed datasets risk widening existing disparities in cardiovascular care when subgroup performance goes unmeasured.

References

- Ahamad, G. N., Shafiullah, Fatima, H., Imdadullah, Zakariya, S. M., Abbas, M., Alqahtani, M. S., & Usman, M. (2023). Influence of optimal hyperparameters on the performance of machine learning algorithms for predicting heart disease. *Processes*, 11(3), Article 734. <https://doi.org/10.3390/pr11030734>
- Alsabhan, W., & Alfadhly, A. (2025). Effectiveness of machine learning models in diagnosis of heart disease: A comparative study. *Scientific Reports*, 15, Article 09423. <https://doi.org/10.1038/s41598-025-09423-y>
- Barreñada, L., Dhiman, P., Liebenberg, E., Collins, G. S., van Smeden, M., Moons, K., & Van Calster, B. (2024). Understanding overfitting in random forest for probability estimation: A visualization and simulation study. *Diagnostic and Prognostic Research*, 8(1), 15. <https://doi.org/10.1186/s41512-024-00177-1>
- Bentejac, C., Csorgo, A., & Martinez-Munoz, G. (2021). A comparative analysis of gradient boosting algorithms. *Artificial Intelligence Review*, 54, 1937-1967. <https://doi.org/10.1007/s10462-020-09896-5>
- Borisov, V., Leemann, T., Sessler, K., Haug, J., Pawelczyk, M., & Kasneci, G. (2024). Deep neural networks and tabular data: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6), 7499-7519. <https://doi.org/10.1109/TNNLS.2022.3229161>
- Chicco, D., & Jurman, G. (2020). Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making*, 20, Article 16. <https://doi.org/10.1186/s12911-020-1023-5>

Fedesoriano. (2021). *Heart failure prediction dataset*. Kaggle.

<https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems (NeurIPS)*, 35. <https://arxiv.org/abs/2207.08815>

Javeed, A., Rizvi, S. S., Zhou, W., Riaz, R., Khan, S. U., & Waqas, S. (2022). Machine learning-based automated diagnostic systems developed for heart failure prediction using different types of data modalities: A systematic review and future directions. *Computational and Mathematical Methods in Medicine*, 2022, 9288452. <https://doi.org/10.1155/2022/9288452>

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), Article 100804. <https://doi.org/10.1016/j.patter.2023.100804>

Lin, W.-C., & Tsai, C.-F. (2020). Missing value imputation: A review and analysis of the literature (2006-2017). *Artificial Intelligence Review*, 53, 1487-1509. <https://doi.org/10.1007/s10462-019-09709-4>

Liu, T., Krentz, A., Lu, L., & Curcin, V. (2025). Machine learning based prediction models for cardiovascular disease risk using electronic health records data: systematic review and meta-analysis. *Eur Heart J Digit Health*, 6(1), 7-22. <https://doi.org/10.1093/ehjdh/ztae080>

Navarro, C. L. A., Damen, J. a. A., Takada, T., Nijman, S. W. J., Dhiman, P., Ma, J., Collins, G. S., Bajpai, R., Riley, R. D., Moons, K. G. M., & Hooft, L. (2022). Completeness of reporting of clinical prediction models developed using supervised machine learning:

a systematic review. *BMC Medical Research Methodology*, 22(1), 12.

<https://doi.org/10.1186/s12874-021-01469-6>

Nusinovici, S., Tham, Y. C., Chak Yan, M. Y., Wei Ting, D. S., Li, J., Sabanayagam, C., Wong, T. Y., & Cheng, C.-Y. (2020). Logistic regression was as good as machine learning for predicting major chronic diseases. *Journal of Clinical Epidemiology*, 122, 56-69. <https://doi.org/10.1016/j.jclinepi.2020.03.002>

Prusty, S., Patnaik, S., & Dash, S. K. (2022). SKCV: Stratified K-fold cross-validation on ML classifiers for predicting cervical cancer. *Frontiers in Nanotechnology*, 4, Article 972421. <https://doi.org/10.3389/fnano.2022.972421>

Raschka, S. (2020). Model evaluation, model selection, and algorithm selection in machine learning. *arXiv preprint*, arXiv:1811.12808v3. <https://arxiv.org/abs/1811.12808>

Teja, M. D., & Rayalu, G. M. (2025). Optimizing heart disease diagnosis with advanced machine learning models: a comparison of predictive performance. *BMC Cardiovascular Disorders*, 25(1), 212. <https://doi.org/10.1186/s12872-025-04627-6>

Theerthagiri, P. (2022). Predictive analysis of cardiovascular disease using gradient boosting based learning and recursive feature elimination technique. *Journal of Information Technology and Digital World*, 4(2), 130-142. <https://doi.org/10.36548/jitdw.2022.2.006>

Tiwari, A., Chugh, A., & Sharma, A. (2022). Ensemble framework for cardiovascular disease prediction. *Computers in Biology and Medicine*, 146, 105624. <https://doi.org/https://doi.org/10.1016/j.compbiomed.2022.105624>

Uddin, S., Khan, A., Hossain, M., & Moni, M. A. (2019). Comparing different supervised machine learning algorithms for disease prediction. *BMC Medical Informatics and Decision Making*, 19. <https://doi.org/10.1186/s12911-019-1004-8>

World Health Organization. (2024). *Cardiovascular diseases (CVDs) fact sheet*. [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))

Xu, Q., Xie, W., Liao, B., Hu, C., Qin, L., Yang, Z., Xiong, H., Lyu, Y., Zhou, Y., & Luo, A. (2023). Interpretability of Clinical Decision Support Systems Based on Artificial Intelligence from Technological and Medical Perspective: A Systematic Review. *Journal of healthcare engineering*, 2023, 9919269. <https://doi.org/10.1155/2023/9919269>

Additional Work:

The six different ways of imputation were assessed systematically and a final method was selected to address the 18.6% missing cholesterol data ($n=171$). The missing data had been confirmed not to be missing at random (in this case, patients with missing values for cholesterol had an 88.3% chance of having heart disease, while patients with values for cholesterol had a 47.7% chance of having heart disease), indicating a clinically meaningful pattern of missing at random data that required careful imputation to address this missing data. The six methods compared were Global Median, Stratified Median (by Age/Sex group), KNN ($k = 5$, without the target variable), KNN ($k = 5$, with the target variable), MICE (Iterative Imputation), and row deletion. As shown in Figure 1A, the imputed cholesterol distributions reveal a clear qualitative difference: Global Median assigns the identical value of 237.0 to all 171 missing entries, producing a sharp artificial spike in the distribution, while KNN (without target) generates 139 unique, patient-personalised estimates that closely mirror the shape of the original non-missing distribution.

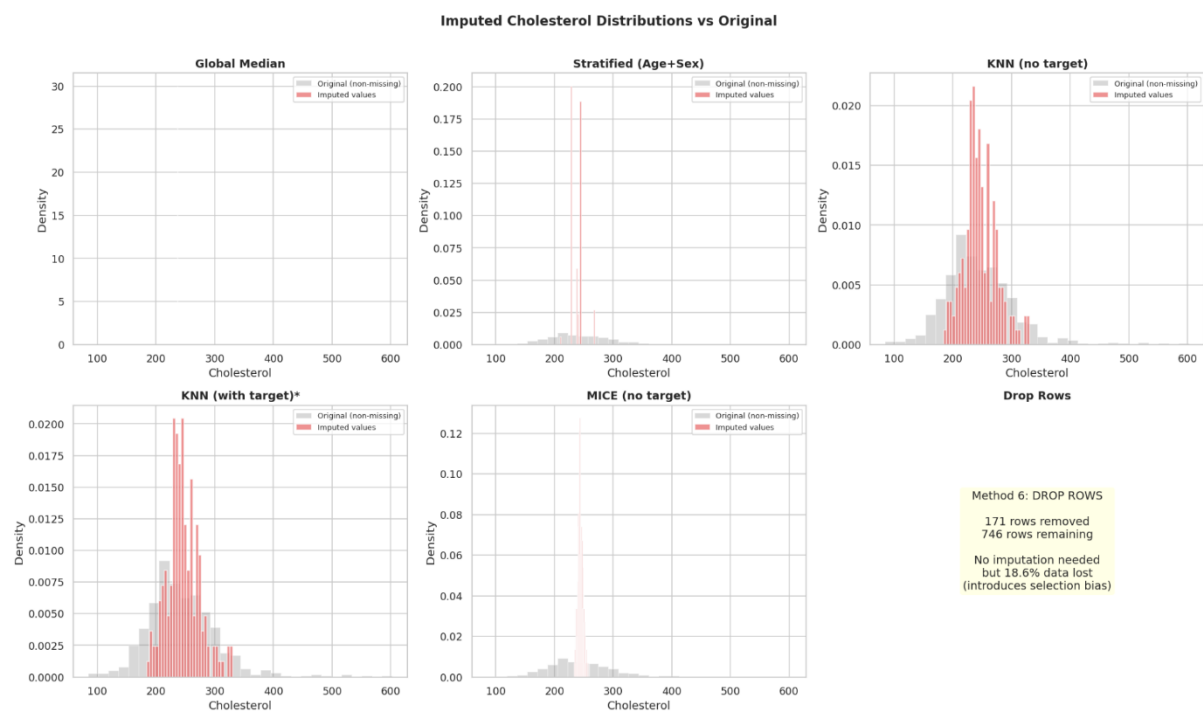


Figure 1A: 6 Subplots of Imputed Cholesterol Distributions Vs. Original

Figure 2A reinforces this finding, demonstrating that KNN (without target) preserves 85% of the original cholesterol–heart disease class separation gap, outperforming both Global Median and MICE in terms of relationship fidelity.

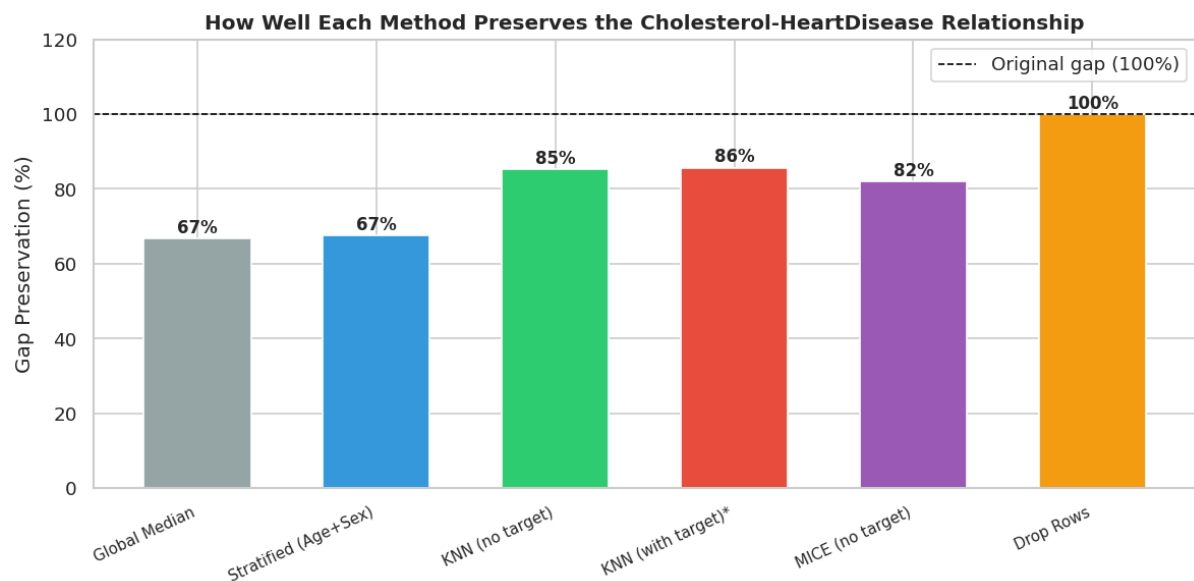


Figure 2A: Bar Chart of Cholesterol–Heart Disease Gap Preservation (%) Per Method

The downstream predictive performance comparison presented in Figure 3A, evaluated using a Random Forest classifier under 10-fold stratified cross-validation across all 917 patients, further confirms this advantage: KNN (without target) achieved the highest F1-score (0.8771) and accuracy (0.8614) among all clinically valid methods.

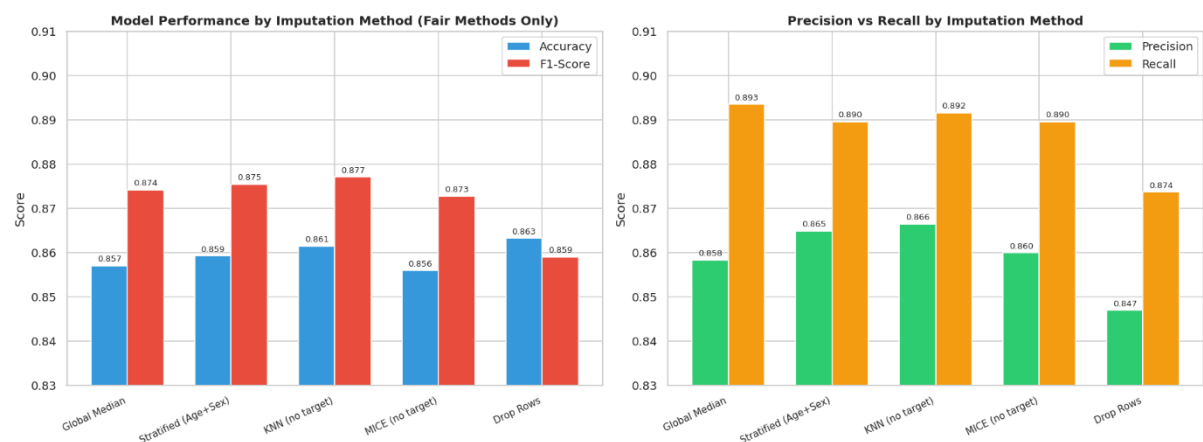


Figure 3A: Accuracy/F1 and Precision/Recall Bars Per Imputation Method

KNN was eliminated as a valid method for making an honest comparison with the target variable because of data leakage. In other words, it would not make any sense to use the HeartDisease label during imputation; therefore, the KNN without the target variable method was finally selected as the imputation method, as it best represents how one would think in a clinical setting: Using a KNN algorithm with only non-target features (age, sex, resting blood pressure, maximum heart rate, ECG) to calculate the cholesterol reading for an individual would involve finding five patients that are most similar to the person for whom we need to perform the calculation and averaging their cholesterol readings to arrive at a reasonable

estimate of what the cholesterol reading would most likely be for the patient in question. The end result of using KNN without a target variable is that all 917 patients remain in the dataset with respect to the cholesterol variable, there is no data leakage, and the value of cholesterol will maintain the statistical structure of the cholesterol variable better than any other analyzed data that was not subject to data leakage.

Dataset Used: heart.csv (Provided with main submission)

Python Code: [KNN.ipynb](#)